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FINAL DEGREE PROJECT

Soyuz 2.1v Liquid Bipropellant Redesign

A Matlab tool for rocket redesign

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Abstract

Soyuz 2.1v is the last born of the Soyuz family launchers and represents the lightweight version of the older *Soyuz 2.1a* and *Soyuz 2.1b*. It is used to quickly insert small payload into Low Earth Orbit resulting in the best launcher choice for military purpose. Since *Soyuz* launch vehicles mainly relied on LOX - RP1 for lower stages propulsion, noting the last trends in propellants configurations, a Soyuz 2.1v first and second stage redesign is accomplished. The stages redesign is done by changing the liquid propellant of each stage and choosing the one which allows to maintain the target mission and to optimize the performance.

Besides the Soyuz 2.1v analysis, a **MATLAB** tool for liquid propellants stages redesign has been developed and validated by the team.

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Nomenclature

Adimensional Quantities

Γ	Vandenkerckhove function	-
γ	Ratio of specific heats $\frac{c_p}{c_v}$	-
λ	Nozzle efficiency	-
ε	Nozzle expansion area ratio $\frac{A_e}{A_t}$	-
C_D	Discharge coefficient	-
C_F	Thrust coefficient	-
L_f	Percentage of the corresponding conical nozzle lenght	-
M	Mach number	-
r	Oxidizer to fuel ratio	-
LF	Fractional Length	-
O/F	Oxidizer to Fuel ratio	-

Physical Quantities

$\dot{m}$	Mass flow rate	kg/s
ρ	Density	kg/m ³
θ	Nozzle divergent angle	deg
a	Speed of sound	m/s
c_p	Isobaric specific heat	J/(kg · K)
D	Diameter	m
g_0	Gravitational acceleration	m/s ²
H	Enthalpy	J
h	Specific enthalpy	J/mol
L	Length	m
m	Mass	kg
M_m	Molar mass	g/mol
P	Pressure	kPa
R	Specific gas constant $\frac{R_u}{M_m}$	J/(kg · K)
R_u	Universal gas constant	J/(mol · K)
T	Temperature	K
t	Time	s
V	Volume	m ³
v	Velocity	m/s
W	Weight	N

z	Altitude	m
A	Cross section area	m^2
I	Impulse	s
L^*	Characteristic length	m
R	Radius	m
T	Thrust	N
t^*	Particle residence time in the combustion chamber	s

Other Symbols

v_n	Expansion efficiency coefficient	-
c	Combustion chamber	-
b	Burning	-
c	Combustion chamber	-
cyl	Cylindrical	-
e	Exit/exhaust	-
f	Fuel	-
i	Initial	-
opt	Optimum expansion	-
out	Outer	-
ox	Oxidizer	-
p	Propellant	-
PL	Payload	-
s	Structural	-
sl	Sea level	-
sp	Specific	-
t	Throat	-
tank	Tank	-
TOT	Total	-
vac	Vacuum	-

1. Introduction

Initially space was the goal to be achieved to show how powerful a nation was and the main purpose of the space race was accelerated by the convergence of lighter nuclear weapons and increased payload capability of ballistic missiles. Nowadays the aim of reaching the space is found mainly in scientific and, only recently, in travelling purposes. Thus, the main goal of rockets design is to minimize gross lift-off weight and increase the payload ratio in order to carry the maximum payload allowable without excessive expenses.

As shown in Tab. 1.3, the first and the second stages of Soyuz 2.1v uses a semi-cryogenic propellant that consists of liquid oxygen (LOX) as oxidizer and RP-1 as fuel, while the third stage uses Dinitrogen tetroxide with Unsymmetrical dimethylhydrazine (N2O4 - UDMH) as propellant. In this analysis, the propellant for each stage (excluding the third one) is changed in order to obtain a different optimization with respect to the real case. The features of the new propellant, like chemical composition and the oxidizer to fuel ratio O/F, are studied through the NASA Chemical Equilibrium with Applications program [NASA CEA](#).

Shown in table 1.1, three of the most commonly used propellants combinations will be evaluated, while the original LOX - RP1 combination will be used for validation sets.

Table 1.1: Evaluated liquid bi-propellant

Acronym	Oxidizer	Fuel
LOX - RP1	O2 - Liquid Oxygen	C ₁₂ H ₂₄ - Semi cryogenic Kerosene
LOX - LH2	O2 - Liquid Oxygen	H ₂ - Cryogenic Hydrogen
N ₂ O ₄ - UDMH	N ₂ O ₄ - Dinitrogen Tetroxide	UDMH - Unsymmetrical Dimethylhydrazine
HNO ₃ - UDMH	HNO ₃ - Nitric Acid	UDMH - Unsymmetrical Dimethylhydrazine

The starting point for the redesign will be the estimated performance of the actual *Soyuz 2.1v* stages, determined from the whole amount of data available about the rocket. Such data are provided both by official ([4]) and unofficial sources and are grouped in Tab. 1.3. Since its military purpose, it is difficult to find all the data needed: that is why T_{s1} has not been considered since very different values for it has been found on online paper. As it can be seen, there is no available data about the combustion chamber pressure P_c , the optimal expansion altitude and exhaust section area available on-line; to compensate for the lack of these two essential parameters, an optimization algorithm has been validated and carried out with the goal of maximize stage performances. The scripts developed could be used for further rocket redesigns. Then, through tank sizing, mass model a discussion about the best propellant configuration will be carried out.

Table 1.2: Original data

PL	2.85 t
ρ_{ox}	1141 kg/m ³
ρ_f	850 kg/m ³
T_{ox}	90.17 K
T_f	266 K

Table 1.3: Original data

	STAGE 1	STAGE 2	STAGE 3
Propellant	LOX - RP1	LOX - RP1	N ₂ O ₄ - UDMH
m_p	119 t	22.8 t	0.5 t
m_s	11 t	2.5 t	0.8 t
D	2.95 m	2.95 m	2.95 m
L	33 m	7 m	3 m
p_c	~	~	~
z_{opt}	~	~	~
A_e	~	~	~
T_{vac}	1910 kN	290 kN	2.9 kN
t_b	250 s	240 s	~
$I_{sp_{sl}}$	291	~	~
$I_{sp_{vac}}$	323	325	307
n main engine	1	1	1
n steering engine	4	0	0

2. Problem Analysis

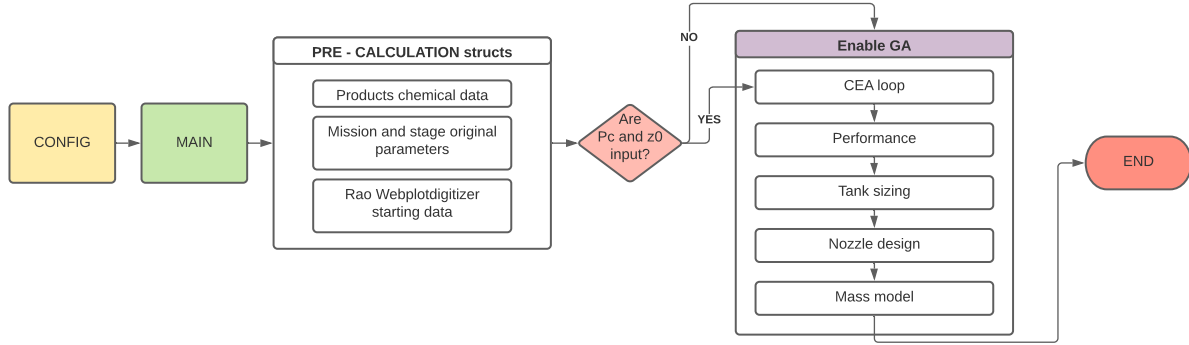


Figure 2.1: Code Workflow

The program workflow is presented in Fig. 2.1. As it is shown the code can work in two different ways: a standard mode where the user can specify combustion chamber pressure P_c and optimal expansion altitude z_{opt} and basing on these two initial parameters, the code will evaluate the best stage and propellant configuration. If P_c and z_{opt} are not known a-priori, the MATLAB Genetic Algorithm solver, GA, will find the best P_c and z_{opt} that accomplish maximum ΔV from Tsiolkovsky equation:

$$\Delta V = I_{sp} g_0 \ln \left(\frac{m_{in}}{m_{fin}} \right) \quad (2.1)$$

Since GA algorithm is computationally expensive, the code needs to simplify the operations contained in the scripts used in the GA algorithm. In order to do this, a pre-calculation section is employed: inside that, a struct¹ of the possible combustion products is created from a file named "rockets.xlsx" which comprehend the coefficients of the *Shomate equation* (Sec. 2.2.1) referred to that element and its molar mass. Furthermore, original mission parameters and target are read beside reading the necessary data for the nozzle design with Rao's method (Sec. 2.4.2).

Inside the GA section, the best O/F is calculated through a continuous use of CEA. Then, using different script shown in 2.3 and 2.4 is possible to obtain the performances, the overall dimensions and the masses of the rocket. This is simplified adding five main assumptions:

1. Evaluated stages have the same vacuum thrust T_{vac} of the original LOX - RP1 stages;
2. Evaluated stages have the same outer diameter D_{out} of the original LOX - RP1 stages;
3. Evaluated stages have the same number of main and steering engines² and of the original stages;
4. Combustion chamber pressure P_c , optimal expansion altitude z_{opt} and O/F shift are the one which maximize ΔV from Tsiolkovsky equation;
5. A 4% performance increment in terms of *total Impulse* I_{TOT} , with respect to the original Soyuz 2.1v I_{TOT} is assumed.

The results showed are the results of an initial run of optimization and are divided into the stage they are referred to. In particular all the following values are obtained from the results of P_c and z_{opt} optimized and are reported in Tab. 2.2.

¹A struct is a composite data type declaration that defines a physically grouped list of variables under one name in a block of memory, allowing the different variables to be accessed via a single pointer.

²In this analysis, steering and main engines P_c and z_{opt} are assumed equal. Therefore, steering and main engines have proportional performance and it is not necessary to treat the firsts separately.

Table 2.1: CEA results (optimized)

STAGE 1	Pc [bar]	z_{opt} [km]	STAGE 2	Pc [bar]	z_0 [km]
LOX - RP1	107.5	15.15	LOX - RP1	96.06	25.10
LOX - LH2	121.7	1.55	LOX - LH2	128.4	24.9
N2O4 - UDMH	126.4	9.19	N2O4 - UDMH	52.0	26.4
HNO3 - UDMH	122.8	13.4	HNO3 - UDMH	101.4	28.5

2.1 Optimization solver

The optimization algorithm used in the process is the Genetic Algorithm (GA), built-in in Matlab. The choice, among the large set of optimization algorithms available, has been done because of the not-regular cost and constraint function over the domain. This implies that a gradient-based solver such as `fmincon`, surrogate gradient or a lot of optimization algorithms is not well suited for the target. In fact, those are optimal once the initial conditions are in the attractive region of the minimum, but it is quite difficult to estimate a guess.

The Genetic Algorithm instead is well suited because of the way it samples the domain and converges to the solution: it does not require a guess but a simple domain. It produces at each macro-iteration a group of samples called population, and according to the cost function and the constraining function will produce the population at the next macro-iteration which is called generation.

For this case, the initial population size has been set to 50 while fitness tolerance up to 10.

2.2 CEA

Table 2.2: CEA results (optimized)

STAGE 1	Pc [bar]	z_{opt} [km]	OF [-]	R [J/(kg K)]	γ [-]	ε [-]
LOX - RP1	107.5	15.15	3.4	295.51	1.1858	70.04
LOX - LH2	121.7	1.55	4.4	771.79	1.1798	16.60
N2O4 - UDMH	126.4	9.19	2.8	343.41	1.2291	33.32
HNO3 - UDMH	122.8	13.4	3	368.20	1.2248	54.60
STAGE 2	Pc [bar]	z_{opt} [km]	OF [-]	R [J/(kg K)]	γ [-]	ε [-]
LOX - RP1	96.06	25.10	3.4	395.8	1.1862	225.19
LOX - LH2	128.4	24.9	5.8	628.54	1.1791	285.50
N2O4 - UDMH	52.00	26.4	3.0	339.89	1.2315	134.79
HNO3 - UDMH	101.4	28.5	3.2	363.32	1.2251	300.97

NASA CEA is a open source program which calculates chemical equilibrium product concentrations from any set of reactants and determines thermodynamic and transport properties for the product mixtures and includes calculations of theoretical rocket performances.

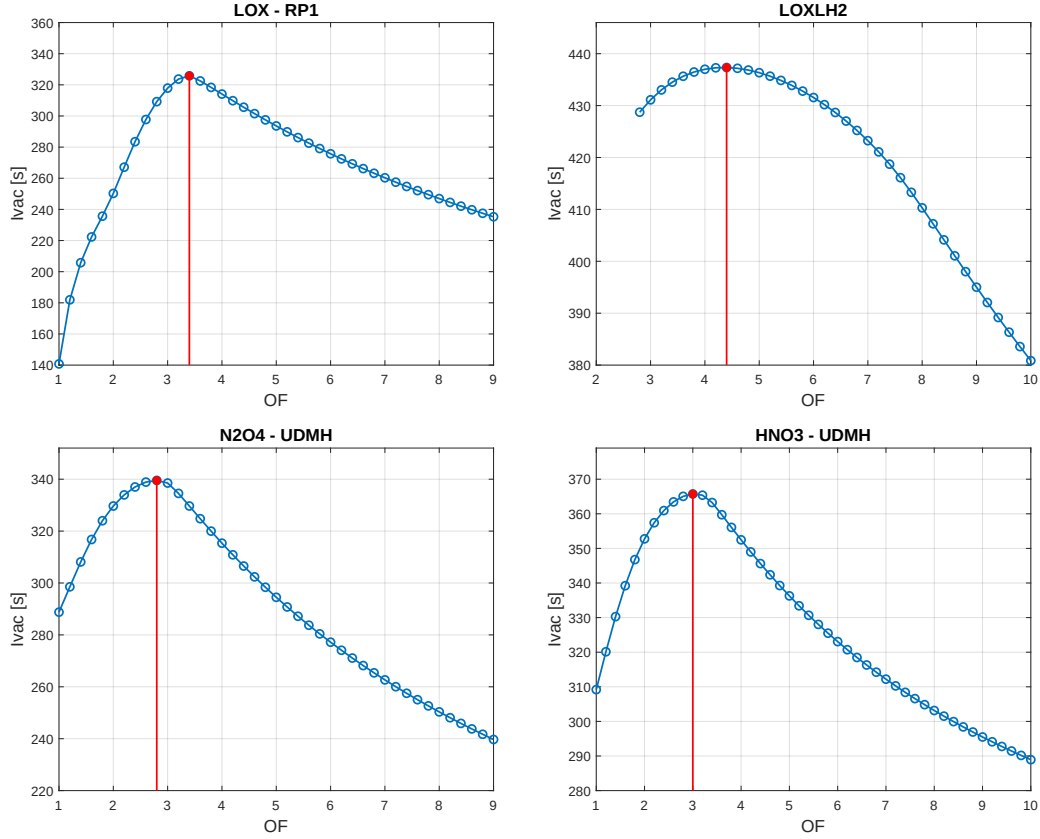


Figure 2.2: Vacuum Specific Impulse trend among the four propellant (First Stage)

After setting the combustion chamber pressure P_c and the optimal expansion altitude z_{opt} , CEA evaluates rocket performances based on different O/F shifts choosing the one which maximizes the vacuum specific impulse $I_{sp,v}$. Studying the evolution of the elements in the combustion chamber through shifting equilibrium is possible to obtain the specific heat of the propellant using a weighted average of the combustion products ones. Fig. 2.2 reports the optimal P_c and z_{opt} for each of the propellant set evaluated (other CEA output data available in App. A.1). Given the i -th product molar mass, it is possible to calculate the i -th product gas constant as it follows:

$$R^i = \frac{R_u}{M_m^i} \quad (2.2)$$

The i -th γ can be calculated by Shomate equation (Sec. 2.2.1) while the average γ can be found with a weighted average:

$$\gamma = \frac{\sum_{i=1}^N \gamma^i M_f^i}{\sum_{i=1}^N M_f^i} = \frac{\sum_{i=1}^N \gamma^i M_f^i}{1} \quad (2.3)$$

where M_f^i is the i -th product mass fraction.

Given the flux specific heat from equation 2.3 the expansion ratio ϵ can be found as follows:

$$\frac{1}{\epsilon} = \left(\frac{\gamma+1}{2} \right)^{\frac{1}{\gamma-1}} \left(\frac{P_e}{P_c} \right)^{\frac{1}{\gamma}} \sqrt{\frac{\gamma+1}{\gamma-1} \left[1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}} \right]} \quad (2.4)$$

The results are reported in Tab. 2.2. In order to obtain more reliable values with the latter epsilons CEA is re-run in order to obtain the optimal O/F used for further studies.

2.2.1 Thermochemical Properties

The *Shomate equation* is adopted with thermochemical properties of both reactants and products to perform CEA calculations. *Shomate equation* approach is an established method to estimate these thermo-chemistry

data using polynomial equations. In particular, isobaric specific heat capacity and specific enthalpy of the i -th compound can be found as it follows:

$$c_{p,i} = A_i + B_i \cdot T + C_i \cdot T^2 + D_i \cdot T^3 + \frac{E_i}{T^2} \quad (2.5)$$

$$h_i = A_i \cdot T + \frac{B_i \cdot T^2}{2} + \frac{C_i \cdot T^3}{3} + \frac{D_i \cdot T^4}{4} - \frac{E_i}{T} + F_i - H_i \quad (2.6)$$

where T is the compound temperature expressed in Kelvin.

There are many polynomial coefficients published in literature depending on the equation adopted for regression and they are all listed in NIST thermochemical property datasheet [5]. Since the particular fuel RP1 used in *Soyuz 2.1v* is not included in CEA libraries³, following CEA User Manual [3], its thermochemical characterization is assumed as a one-formula surrogate fuel (C12H24), using the fitting coefficients taken from [14]. Using Eq. 2.6 evaluated at $T = 266 \text{ K}$ ⁴ yields:

$$H_{RP1} = -2.253736 \cdot 10^6 \text{ [J/mol]}$$

which was used in LOX-RP1

2.3 Performance

It is convenient to evaluate Vandekerckhove function based on the optimal γ :

$$\Gamma = \sqrt{\gamma \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}}} \quad (2.7)$$

The exhaust velocity can be found as the product between the theoretical limited velocity v_{lim} and the expansion efficiency coefficient v_n :

$$v_e = \sqrt{1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}}} \cdot \sqrt{\frac{2\gamma}{\gamma-1} RT_c} \quad (2.8)$$

While the expansion efficiency coefficient v_n is obtained as follow:

$$v_n = \sqrt{1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma-1}{\gamma}}} \quad (2.9)$$

The vacuum thrust coefficient is a measure of vacuum thrust increase due to the divergent region, normalized with reference to the static thrust at throat section. It can be expressed as it follows:

$$C_F^{vac} = \eta_{is} \left(\Gamma v_n \sqrt{\frac{2\gamma}{\gamma-1}} + \left(\frac{P_e}{P_c} \right) \varepsilon \right) \quad (2.10)$$

where a nozzle isentropic expansion efficiency of 0.97 has been considered.

Furthermore the throat section area in vacuum conditions can be found as:

$$A_t = \frac{T_{vac}}{P_c C_F^{vac}} \quad (2.11)$$

Thus, exhaust section area is given by:

$$A_e = \varepsilon A_t \quad (2.12)$$

³CEA works with a libraries file, called *thermo.lib*.

⁴Soyuz 2.1v stages has the capability of *sub-cooling* the RP1 fuel.

The mass flow rate of propellant, through the combustion chamber:

$$\dot{m}_p = \frac{T_{vac} - P_e A_e}{v_e} \quad (2.13)$$

If the rocket is provided with a gas generator system, an additional 4% for the mass flow rate will be considered.

The overall engine specific impulse is then found as:

$$I_{sp}^{vac} = \eta_{is} \left(\frac{T_{vac}}{\dot{m}_p g_0} \right) \quad (2.14)$$

Table 2.3: Performances results (optimized)

STAGE 1	Γ	v_e [m/s]	$C_F^{vac} [-]$	A_t [m ²]	A_e [m ²]	$\dot{m}_p$ [kg/s]	I_{sp}^{vac} [s]
LOX - RP1	0.6458	2865.2	1.907	0.093	6.529	639.825	322
LOX - LH2	0.6446	4100.7	1.7804	0.088	1.460	435.864	473
N2O4 - UDMH	0.6541	2964.6	1.8036	0.084	2.786	616.170	334
HNO3 - UDMH	0.6533	3192.5	1.8436	0.084	4.600	575.803	358

Table 2.4: Performances results (optimized)

STAGE 2	Γ	v_e [m/s]	$C_F^{vac} [-]$	A_t [m ²]	A_e [m ²]	$\dot{m}_p$ [kg/s]	I_{sp}^{vac} [s]
LOX - RP1	0.6458	3008.1	1.9827	0.015	3.429	93.586	334
LOX - LH2	0.6444	4595.9	2.0086	0.011	3.231	61.306	510
N2O4 - UDMH	0.6546	3087.5	1.8893	0.030	3.980	91.333	342
HNO3 - UDMH	0.6534	3367.2	1.9356	0.015	4.446	84.189	372

2.4 Rocket Design

2.4.1 Tank Sizing

Knowing the mission of the rocket and its performances is possible to obtain the mass of the propellant required in order to reach the target. Different propellant results in different conclusion. Generally is possible to implement the tank sizing starting from the total impulse, the combustion time and the oxidizer to fuel ratio to obtain the volume of the propellant. Adding a generic 4% performance increase:

$$I_{tot,real} = 1.04 \cdot I_{tot} \quad (2.15)$$

$$\dot{m}_{ox} = \frac{r}{1+r} \cdot \dot{m}_p \quad (2.16)$$

$$\dot{m}_f = \frac{1}{1+r} \cdot \dot{m}_p \quad (2.19)$$

$$m_{ox} = 1.06 \cdot \dot{m}_{ox} \cdot t_b \quad (2.17)$$

$$m_f = 1.06 \cdot \dot{m}_f \cdot t_b \quad (2.20)$$

$$V_{ox} = \frac{m_{ox}}{\rho_{ox}} \quad (2.18)$$

$$V_f = \frac{m_f}{\rho_f} \quad (2.21)$$

The extra propellant is stored in order to avoid problem linked to the non-instantaneous combustion and because the propellant has to reach the combustion chamber through different ducts. Problem of these types can be evaporation, vaporization and residual propellant trapped in valves or in the tank itself. In order to avoid them it was used a coefficient of 6% for an extra propellant mass (Eq.2.17, Eq.2.20).

Fuel and oxidizer are contained in two tanks which can be combined in different configurations. Thus the space used can vary and the most common is the tandem configuration where the two tanks are one above the other one and the total space will be the sum of the lengths of the two tanks (particular attention is to pay to the ducts required for the passage of the propellant which increases the total diameter of the rocket). In addition the tanks used for cryogenic propellant are not completely filled in order to allow the expansion of the liquid propellant and for this reason the propellant volume is not equal to the tank volume. In particular it is considered an extra 3%:

$$V_{ox} \neq V_{ox}^t \quad (2.22)$$

$$V_f \neq V_f^t \quad (2.24)$$

$$V_{ox}^{tank} = 1.03 \cdot V_{ox} \quad (2.23)$$

$$V_f^{tank} = 1.03 \cdot V_f \quad (2.25)$$

Considering two cylindrical tanks for the fuel and the oxidizer:

$$V_{cyl} = \pi R^2 \cdot L = \pi \frac{D^2}{4} \cdot L \quad (2.26)$$

The diameter D of the Soyuz 2 is univocally determined as $D \approx 2.5m$.

$$V_{ox}^{tank} = \pi \frac{D^2}{4} \cdot L_{ox} \quad (2.27)$$

$$V_f^{tank} = \pi \frac{D^2}{4} \cdot L_f \quad (2.29)$$

$$L_{ox} = \frac{4V_{ox}^t}{\pi D^2} \quad (2.28)$$

$$L_f = \frac{4V_f^t}{\pi D^2} \quad (2.30)$$

The final length will be the sum of the oxidizer and fuel lengths:

$$L_{tot} = L_{ox} + L_f \quad (2.31)$$

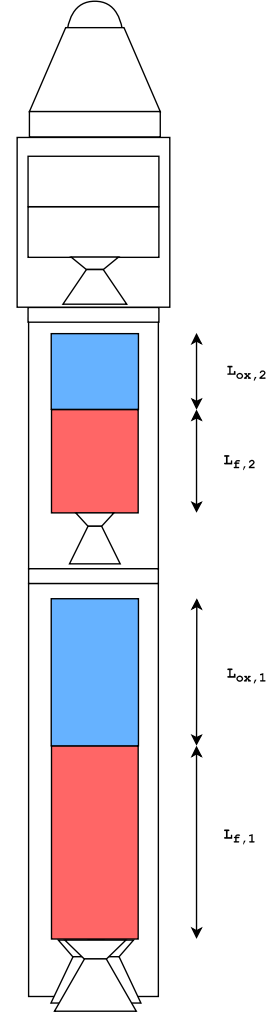


Table 2.5: Performances results (optimized)

STAGE 1	$I_{tot}^{real}[Ns]$	$\dot{m}_{ox}[kg/s]$	$\dot{m}_f[kg/s]$	$m_{ox}[t]$	$m_f[t]$	$V_{ox}[m^3]$	$V_f[m^3]$	$L_{ox}[m]$	$L_f[m]$	$L_{tot}[m]$
LOX - RP1	377066970	524.075	154.140	103.46	30.43	93.4	36.87	18.3	7.22	29.14
LOX - LH2	392149648	376.457	85.558	77.29	17.57	69.77	255.38	13.7	50	65
N2O4 - UDMH	392149648	481.261	171.879	98.81	35.29	70.68	45.84	13.8	8.9	25.1
HNO3 - UDMH	119793	457.764	152.588	93.99	31.33	64.11	40.70	12.6	8.0	23.5
STAGE 2	$I_{tot}^{real}[s]$	$\dot{m}_{ox}[kg/s]$	$\dot{m}_f[kg/s]$	$m_{ox}[kg]$	$m_f[kg]$	$V_{ox}[m^3]$	$V_f[m^3]$	$L_{ox}[m]$	$L_f[m]$	$L_{tot}[m]$
LOX - RP1	72692100	76.655	22.546	19.22	5.65	17.3	6.8	3.4	1.3	7.6
LOX - LH2	75599784	55.428	9.557	14.45	2.5	13.1	36.2	2.6	7.1	12.4
N2O4 - UDMH	75599784	72.610	24.203	18.93	6.31	13.5	8.2	2.7	1.6	7.2
HNO3 - UDMH	75599784	68.000	21.248	17.73	5.54	12.1	7.2	2.4	1.4	6.9

2.4.1.1 Combustion Chamber

The combustion chamber must have adequate dimensions so the propellants can react until the equilibrium and be as small as possible to reduce the weight. In addition, the necessary time and space depend on the injectors performance. Following [1], considering the volume of a combustion chamber V_c where the resultants propellants and the propulsive gas are averagely set to a density ρ_c , the gas mass will be:

$$m_c = \rho_c V_c \quad (2.32)$$

knowing the mass flow rate of gas through the combustion chamber $\dot{m}$ it is possible to calculate how much time a particle remains in the combustion chamber:

$$t_c = \frac{\rho_c}{\dot{m}_p} \quad (2.33)$$

The chemical equilibrium is reached after a definite amount of time which is called t^* and can be used to calculate the volume:

$$t_c \geq t^* \Rightarrow V_c \geq \frac{\dot{m}_p t^*}{\rho_c} \quad (2.34)$$

Considering the gas in the combustion chamber an ideal gas and the temperature and the pressure equal to the ones in the nozzle the volume can be rewritten as:

$$V_c \geq \frac{\Gamma P_c A_t t^*}{\rho_c \sqrt{RT_c}} = \frac{\Gamma P_c A_t RT_c t^*}{P_c \sqrt{RT_c}} = \Gamma A_t t^* \sqrt{RT_c} \quad (2.35)$$

Introducing the length $L_c = V_c / A_t$, index of the axial dimension of the combustor as:

$$L_c = \frac{V_c}{A_t} \geq \Gamma A_t t^* \sqrt{RT_c} \quad (2.36)$$

It is possible to obtain the characteristic length L^* which expresses the minimum axial dimension of the combustion chamber in order to allow the chemical reactions:

$$L_c \geq L^* \Rightarrow L^* = t^* \Gamma \sqrt{RT_c} \quad (2.37)$$

$$a_c = \sqrt{\gamma RT_c} \Rightarrow v_c = M_c a_c \quad (2.38)$$

$$A_c = \frac{\dot{m}_p}{\rho_c v_c} \Rightarrow R_c = \sqrt{\frac{A_c}{\pi}} \quad (2.39)$$

$$L^* = \frac{V_c}{A_t} \quad (2.40)$$

TABLE 4-1.—Recommended Combustion Chamber Characteristic Length (L^*) for Various Propellant Combinations

Propellant combination	Combustion chamber characteristic length (L^*), in.
Chlorine trifluoride/hydrazine-base fuel	30-35
Liquid fluorine/hydrazine	24-28
Liquid fluorine/liquid hydrogen (GH_2 injection)	22-26
Liquid fluorine/liquid hydrogen (LH_2 injection)	25-30
Hydrogen peroxide/RP-1 (including catalyst bed)	60-70
Nitric acid/hydrazine-base fuel	30-35
Nitrogen tetroxide/hydrazine-base fuel	30-35
Liquid oxygen/ammonia	30-40
Liquid oxygen/liquid hydrogen (GH_2 injection)	22-28
Liquid oxygen/liquid hydrogen (LH_2 injection)	30-40
Liquid oxygen/RP-1	40-50

Figure 2.3: Recommended Combustion Chamber Characteristic Length

2.4.2 Nozzle Design

The nozzle design for a rocket is important and must be studied in order to obtain the correct evaluation of the generated thrust. In fact this one is affected by a corrective factor λ that represents the two dimensional losses of the flow divergence, resulting in a thrust loss along the nozzle longitudinal axis. It has values typically over 90%. To minimize this effect is therefore necessary to limit the half-open angles of the divergent or to realize a nozzle with variable angle.

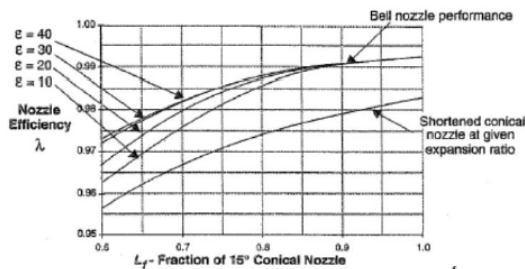


Figure 2.4: Variation of bell nozzle efficiency with length ratio and expansion ratio.

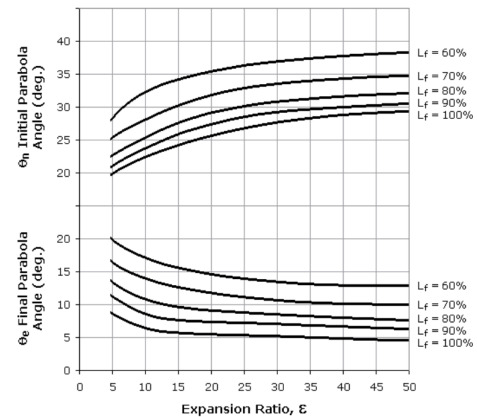


Figure 2.5: θ_i and θ_e as function of expansion area ratio.

The approach chosen to design the nozzle is the Rao's Approximation as in [2]. It involves a procedure which is aimed to find the most important features of the nozzle: λ , l_f and the angles which will be used to design its two-dimensional shape θ_i and θ_e . The bell nozzle made through this method is based on the exploitation of the high pressure in the throat section, which allows to impose a wide angle of divergence that decreases along the axis. The Fig.2.4 represents the trend of ϵ linked to λ and l_f . Thus, the design of the nozzle can be done in two different ways:

- Determine λ , knowing ϵ and l_f ;
- Determine l_f , knowing λ and ϵ .

Since λ (i.e. η_{is}) is set in Sec. 2.3, the latter way is adopted. ϵ is given, therefore the corresponding value of l_f is found using the relationship shown in Fig.2.4 where l_f represents the percentage of divergent part length compared to a conical divergent with $\theta_i = 15^\circ$. Considering the graph (Fig.2.4) it is shown that the lines flatten correspond to $\epsilon = 40$. In those cases when $\epsilon > 40$, it is reasonable to assume the maximum value.

In the graph the curves represented referred to $\varepsilon = 10, 20, 30, 40$, if the ε is a different value an interpolation will be necessary as shown in:

In the first image it has been used a polynomial interpolant (respectively of fourth degree for x , l_f and of first degree for the y , ε), instead for the second one, the inverse function, a linear interpolant solver has been adopted (Fig.2.6). In order to achieve such a purpose, the Matlab function `fit` has been employed.

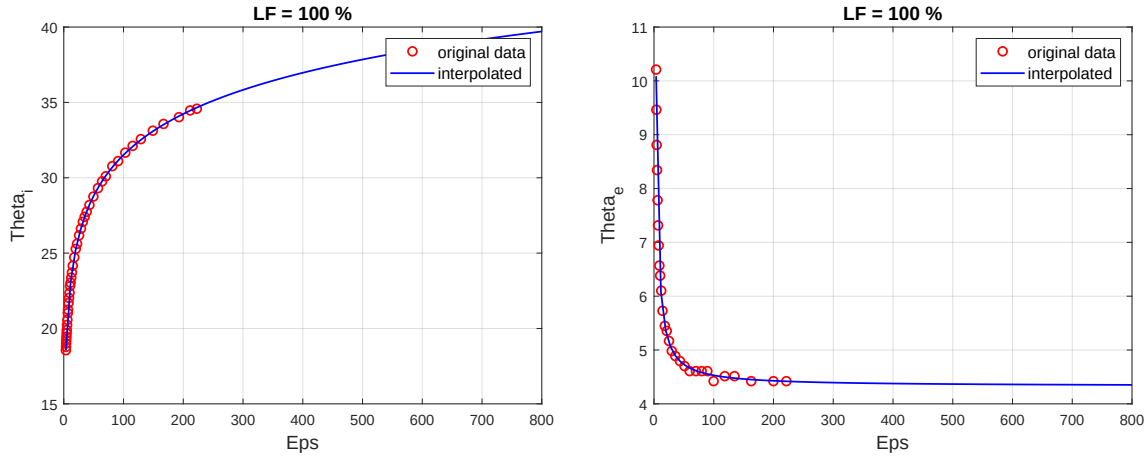


Figure 2.6: Data fitting

After the corresponding value of λ is found, θ_i and θ_e must be evaluated. In this case the maximum value shown in figure is $\varepsilon = 50$ which is not correct for a generic nozzle design. In fact, greater ε values were calculated (Tab 2.2). In Fig. 2.4 a first polynomial interpolation is created to increase the range of values which are needed to create the 3D plots (Fig.2.7). Furthermore the 3D plots of Fig.2.5 are elaborated in Fig.2.8.

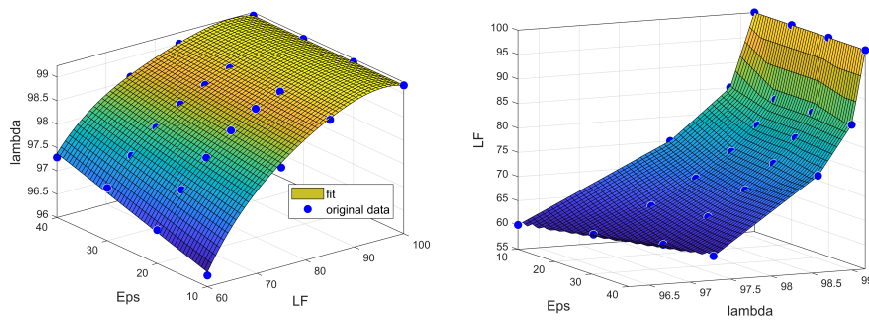


Figure 2.7: Variation of bell nozzle efficiency with length ratio and expansion ratio

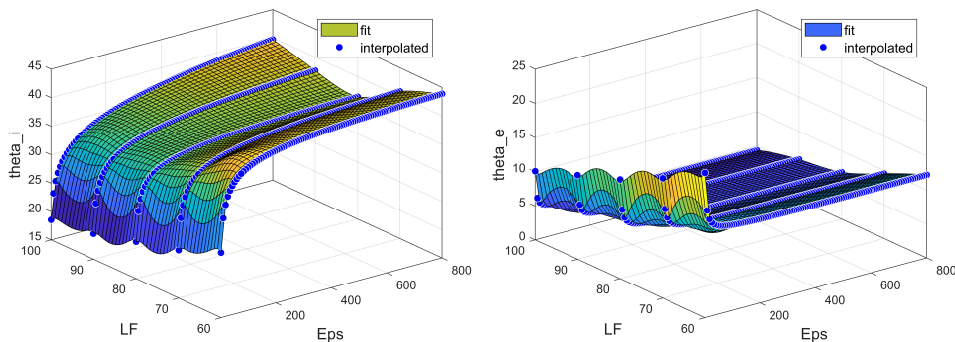


Figure 2.8: θ_i and θ_e as function of expansion area ratio.

For the shape of the nozzle, it is implemented Rao's parabolic approximation of the nozzle contour, which stands on the geometric assumptions in (Fig.2.9) explained as follow: the curve of the section just before the throat section has a radius equal to $1.5R_t$, with R_t as radius of throat; the second curve that links the previous one to the divergent parabolic section has a radius equal to $0.382 R_t$. Eventually, the divergent portion is represented by a parabola starting at an angle of θ_i and ending at an angle of θ_e .

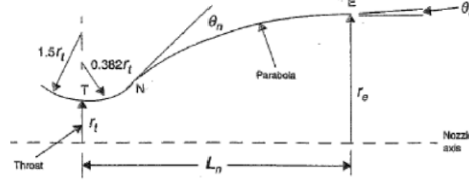


Figure 2.9: Rao's geometric assumptions

Once the total length of the diverging section is calculated (Eq.2.41), the contour of the nozzle is obtained through a geometric approach.

$$L_{bell} = l_f \frac{R_t (\sqrt{\varepsilon} - 1)}{\tan 15^\circ} \quad (2.41)$$

In particular the equation of the parabolic part is:

$$x = ay^2 + by + c \quad (2.42) \quad \frac{dx}{dy} = \frac{1}{\tan \theta} \quad (2.43)$$

To determine the parabola coefficients it's necessary to resolve the derivative of the equation at the point where the curve from the throat meets the beginning of the parabola. Then, the initial and final values of the parabolic section are replaced into the equation. Operating in this way, a linear system is obtained in matrix form that can be solved for the coefficients:

$$\begin{cases} \frac{dx}{dy} &= 2ay + b \rightarrow 2ay_i + b = \frac{1}{\tan \theta_i} \\ x_e &= ay_e^2 + by_e + c \rightarrow aR_e^2 + bR_e + c = L_{bell} \\ ay_i^2 + by_i + c &= x_i \end{cases} \quad (2.44)$$

$$\begin{bmatrix} 2y_i & 1 & 0 \\ R_e^2 & R_e & 1 \\ y_i^2 & y_i & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \frac{1}{\tan \theta_i} \\ L_{bell} \\ x_i \end{bmatrix} \quad (2.45)$$

The data in the following table shown the features of the nozzles for each configuration:

	LOX/RP1	LOX/LH2	N2O4/UDMH	HNO3/UDMH
Bell length	1.26	0.5336	0.7736	1.0383
Throath area	0.015	0.011	0.030	0.015
Exit area	3.429	3.231	3.980	4.446
ε	225.19	285.50	134.79	300.97

2.5 Mass Model

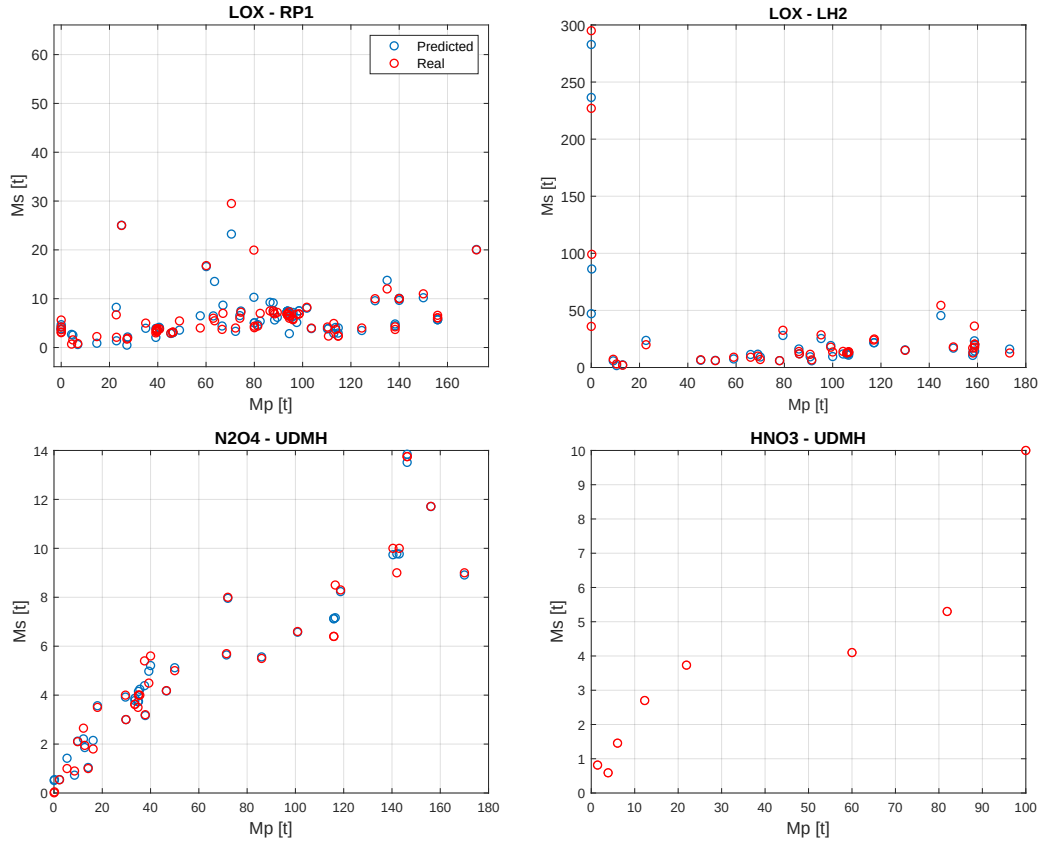


Figure 2.10: Predicted vs Real values

A mass model has been developed to estimate the stage structural mass. The starting point is a set of modern rockets all listed in Appendix C.2.1. A polynomial interpolation can be performed among all those data in order to create an interpolant function which, given an input, can return the stage structural mass. To achieve such a goal, it has been chosen a second order polynomial. The input consists of six variables and can be seen in Tab. 2.6.

Table 2.6: Mass Model input variables

	x_1	x_2	x_3	x_4	x_5	x_6
variable	m_p	I_{vac}	n_{engine}	L	D	T_{vac}

All the input variables are calculated in the previous section of the algorithm as it follows:

- Propellant mass m_p : calculated in Tank sizing section (Sec. 2.4.1);
- Vacuum Specific Impulse I_{vac} : calculated in CEA section (Sec. 2.2);
- Number of main engines n_{engine} : equal to original stage one by hypothesis;
- Stage length L : calculated in Tank sizing section (Sec. 2.4.1) and Nozzle design (Sec. 2.4.2) sections;
- Stage Diameter D : equal to original stage one by hypothesis;
- Vacuum thrust m_p : equal to original stage one by hypothesis;

Table 2.7: Mass Model validation sets

propellant	stage	$M_{p,real}$	$M_{s,real}$	I_{vac}	n° engines	length	Diameter	T_{vac}	$M_{s,pred}$	$ M_{s,real} - M_{s,pred} $
LOX - RP1	Falcon 9 - 2nd stage [11]	115.3 t	4.5 t	355 s	1	18 m	3.66 m	935 kN	5.1256 t	0.6256 t
	Soyuz FG - 3rd stage [13]	22.8 t	2.8 t	359 s	1	6.7 m	2.66 m	294 kN	3.367 t	0.567 t
LOX - LH2	Ariane 5 - 1st stage [8]	158.1 t	12.7 t	425 s	1	30.5 m	5.4 m	1105.3 kN	12.7777 t	0.7777 t
	Ariane 1 - 3st stage [6]	8.23 t	1.457 t	443 s	1	10.5 m	2.66 m	61.67 kN	5.847 t	4.39 t
N2O4 - UDMH	Ariane 44LP - 2nd stage [7]	228 t	35 t	270 s	1	27.6 m	3.8 m	700 kN	14.8 t	3.09 t
	Chang Zheng 2C - 2nd stage [10]	8.23 t	4 t	295 s	1	7.5 m	3.35 m	762 kN	3.96 t	0.039 t
HNO3 - UDMH	Kosmos 65S3 - 1st stage [12]	81.9 t	12.7 t	292 s	1	22.4 m	2.4 m	1740 kN	5.282 t	0.282 t
	Atlas Able - 2nd stage [9]	2.268 t	0.816 t	270 s	1	6.57 m	0.84 m	34.3 kN	0.816 t	0.016 t

The fundamental guideline to build an interpolant function is the minimization of polynomial degree and the number of polynomial coefficients; in fact, since the code adopted a GA optimization solver, it is essential to minimize computational cost in order to speed up the algorithm. The code performs a pre-calculation parsing to obtain better interpolated values starting from a modest number of coefficients.

Given the original stage propellant mass and the rockets examples (App. C.2.1) the interpolant function will be based on the ones that satisfy:

$$m_p \leq 1.5 \cdot m_p \quad (2.46)$$

This allows the function to be easier to use and to return more accurate predicted values. The function considered is a second order degree polynomial one, resulting in 28 polynomial coefficients:

$$f(\underline{x}) = a_0 + \sum_{i=1}^6 a_i \cdot x_i + \sum_{i=1}^5 \left(\sum_{j=i+1}^6 a_{(-\frac{1}{2}i^2 + \frac{11}{2}i+j)} \cdot x_i x_j \right) + \sum_{i=1}^6 a_{i+21} \cdot x_i \quad (2.47)$$

In order to find the best coefficients set, it is used the MATLAB in-built function `lsqcurvefit`⁵. This function can solve non-linear least squares problems; it implements two different algorithms: trust region reflective and Levenberg-Marquardt. The latter is the one used in the project. `lsqcurvefit` needs an input guess, $\underline{a}_0$, and finds the best coefficients $\underline{a}$ to best fit the nonlinear functions defined in Eq. 2.47.

The set of optimal coefficients can be seen in App. C.1 drafted respectively for each evaluated stage and evaluated propellant.

2.5.1 Validation

In order to evaluate the correctness of the mass model, the interpolant function defined in Eq. 2.47 has been used on a sets of existing rockets which are not included in the starting example upon which the model is created. In particular, two stages for each propellant type configuration has been used for such a validation. As it can be seen in Tab. 2.7, the errors made are relatively small. Thus, the assumptions made and the interpolant function defined in Eq. 2.47 are suitable for the purpose of the model created.

2.6 Model Validation

In order to validate the implemented model, attempts have been made to maximize the performances of the existing stages with the aim of check if the final results could match to the real ones. Recalling that the following variables has been kept identical to the ones in the original stages:

- diameter
- T_{vac}
- engine number

⁵<https://www.mathworks.com/help/optim/ug/lsqcurvefit.html>

Table 2.8: Comparison between original and predicted stages

	STAGE 1 ORIGINAL	STAGE 1 PREDICTED	STAGE 2 ORIGINAL	STAGE 2 PREDICTED
Propellant	LOX - RP1	LOX - RP1	LOX - RP1	LOX - RP1
m_p	119 t	126.3 t	22.8 t	23.5 t
m_s	11 t	11.9 t	2.8 t	5.9 t
D	2.95 m	2.95 m	2.95 m	2.95 m
L	33 m	29.14 m	3 m	7.6 m
P_c	~	107.45 bar	~	96.06 bar
z_{opt}	~	15.15 km	~	25.10 km
A_e	~	6.53 m ²	~	3.43 m ²
T_{vac}	1910 kN	1910 kN	290 kN	290 kN
t_b	250 s	197 s	240 s	250 s
$I_{sp_{vac}}$	323 s	322 s	325 s	334 s
n main engine	1	1	1	1
n steering engine	4	0	0	0

In contrast to the previous assumptions, the 4% performance increase of I_{tot} is no longer needed for the model validation. The results are presented in Tab.2.8. Leaving out the discrepancy in t_b values, the predicted results are comparable to the original ones. Thus, the model can be used to modify the propellants of a real rocket to create a new one with different performance.

4. Conclusion

It is possible now to present the main results obtained during the process adding the Tsiolkovskij equation, given by:

$$\Delta V = I_{sp} g_0 \ln \left(\frac{m_{in}}{m_{fin}} \right) \quad (3.1)$$

Stage 1 - PROP	ΔV	Stage 2 - PROP	ΔV
LOX - RP1	3933.9 m/s	LOX - RP1	5550.5 m/s
LOX - LH2	5270.5 m/s	LOX - LH2	4752.9 m/s
N2O4 - UDMH	4450.3 m/s	N2O4 - UDMH	4734.2 m/s
HN03 - UDMH	4925.0 m/s	HN03 - UDMH	5014.8 m/s

Table 3.1: Stage 1 results

	STAGE 1 ORIGINAL	STAGE 1 PREDICTED	STAGE 1 PREDICTED	STAGE 1 PREDICTED
Propellant	LOX - RP1	LOX - LH2	N2O4 - UDMH	HN03 - UDMH
m_p	119 t	89.5 t	126.5 t	118.2 t
m_s	11 t	3.7 t	5.5 t	1.57 t
D	2.95 m	2.95 m	2.95 m	2.95 m
L	33 m	65 m	25.1 m	23.5 m
P_c	~	121.7 bar	126.4 bar	122.8 bar
z_{opt}	~	1.55 km	9.19 km	13.4 km
A_e	~	1.46 m ²	2.79 m ²	4.6 m ²
T_{vac}	1910 kN	1910 kN	1910 kN	1910 kN
t_b	250 s	205 s	205 s	205 s
$I_{sp_{vac}}$	323 s	473 s	334 s	358 s
n main engine	1	1	1	1
n steering engine	4	4	4	4

The above tables show different ΔV for each propellant and for each stage. As can be deduced, the model predicts a stage 1 which uses LOX-LH2 with better performance than the original one which uses LOX-RP1. In particular the use of the new propellant permits to obtain a greater value of ΔV even though the size of the rocket is bigger because the cryogenic propellant needs a different and stronger structure. Studying the second stage the most suitable propellant is LOX-RP1 which has a bigger ΔV . In addition the second stage is generally smaller and must be lighter than the first one in order to reach the target mission. Theoretically, it is possible to change the propellant of a single stage maintaining LOX-RP1 for the second stage and changing the propellant in the first stage to LOX-LH2, but having two separate rocket engines consists in more problems from an engineering perspective such as more simulation required and the use of different materials resulting

Table 3.2: Stage 2 results

	STAGE 2 ORIGINAL	STAGE 2 PREDICTED	STAGE 2 PREDICTED	STAGE 2 PREDICTED
Propellant	LOX - RP1	LOX - LH2	N2O4 - UDMH	HN03 - UDMH
m_p	22.8 t	15.9 t	23.8 t	21.9 t
m_s	2.8 t	4.4 t	2.1 t	2.0 t
D	2.95 m	2.95 m	2.95 m	2.95 m
L	3 m	12.4 m	7.2 m	6.9 m
P_c	~	128.4 bar	52.0 bar	101.4 bar
z_{opt}	~	24.9 km	26.40 km	28.5 km
A_e	~	3.23 m ²	4.00 m ²	4.45 m ²
T_{vac}	290 kN	290 kN	290 kN	290 kN
t_b	240 s	260.7 s	260.7 s	260.7 s
I_{spvac}	325 s	510 s	342 s	372 s
n main engine	1	1	1	1
n steering engine	0	0	0	0

in a more expansive project. A different approach could be changing the propellant of all stages to LOX-LH2 which can result in better performances, but originally the Soyuz 2.1v from a soviet rocket launcher which has always used LOX-RP1 since his original development during the cold war.

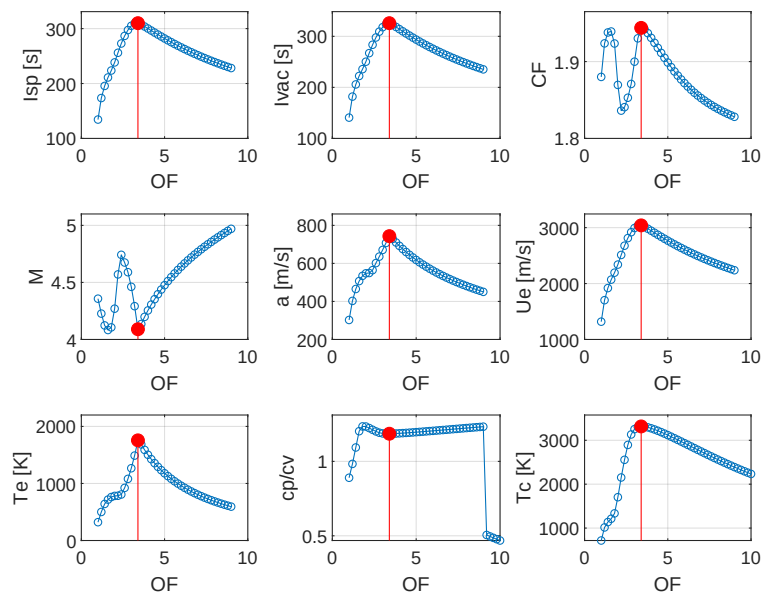
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Appendix A

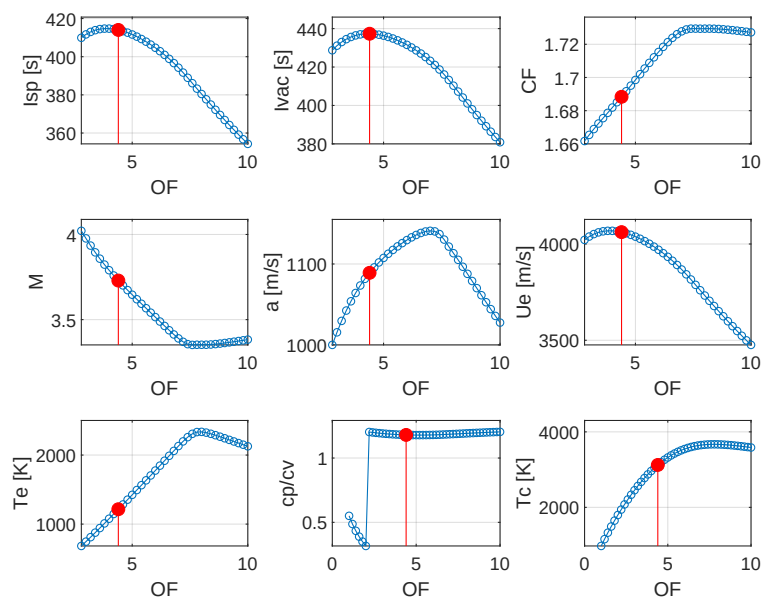
CEA

A.1 CEA Parameters (optimized)

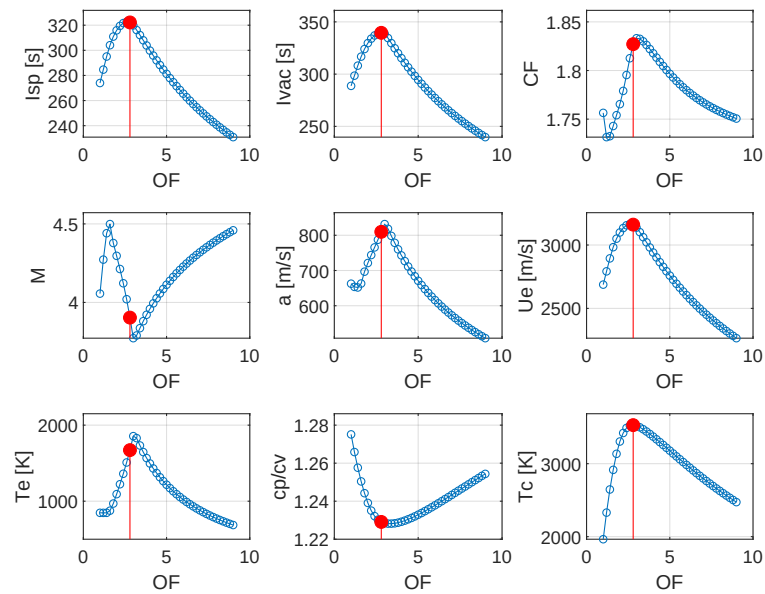
A.1.1 LOX - RP1



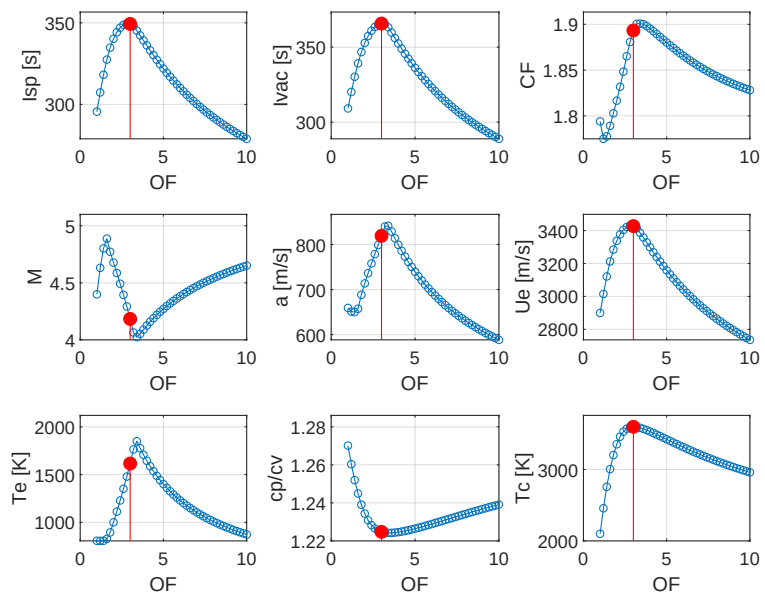
A.1.2 LOX - LH2



A.1.3 N2O4 - UDMH



A.1.4 HN03 - UDMH



A.2 Combustion Products Characterization (optimized)

Table A.1: LOX - RP1 products mass fractions

Product	CO	CO2	H	H2O	H2	H2O	H2O2	O
Mass fraction %	0.430	38.544	0.001	0.025	0.004	15.444	0.002	0.141

Product	OH	O2
Mass fraction %	1.053	44.355

Table A.2: LOX - LH2 products mass fractions

Product	H	H2O	H2	H2O	H2O2	O	OH	O2
Mass fraction %	0.198	0.030	1.965	85.089	0.008	0.875	8.594	3.240

Table A.3: N2O4 - UDMH products mass fractions

Product	CO	CO2	H	HN2O	H2O	H2	H2O	H2O2
Mass fraction %	0.299	17.838	0.001	0.003	0.021	0.006	14.348	0.002

Product	NO	NO2	N2	N2O	O	OH	O2
Mass fraction %	2.850	0.037	31.123	0.002	0.152	1.079	32.240

Table A.4: Nitric Acid - UDMH products mass fractions

Product	CO	CO2	H	HNO	HN2O	H2O	H2	H2O
Mass fraction %	0.299	17.838	0.001	0.003	0.021	0.006	14.348	0.002

Product	H2O2	NO	NO2	N2	N2O	O	OH	O2
Mass fraction %	2.850	0.037	31.123	0.002	0.152	1.079	32.240	0.002

Appendix B

Thermochemical Properties

product	A	B	C	D	E	Mm
O2	20.91	10.72	-2.02	0.15	9.25	0.0319988
O	20.79	0.00	0.00	0.00	0.00	0.0159994
H2O2	34.26	55.18	-35.15	9.09	-0.42	0.0340147
H2O	41.96	8.62	-1.50	0.10	-11.16	0.0180153
H2	43.41	-4.29	1.27	-0.10	-20.53	0.00201588
HO2	45.88	8.81	-1.64	0.10	-10.17	0.0330067
OH	28.75	4.71	-0.81	0.05	-2.75	0.0170073
H	20.79	0.00	-0.00	0.00	0.00	0.00100794
CO	35.15	1.30	-0.21	0.01	-3.28	0.02801
CO2	58.17	2.72	-0.49	0.04	-6.45	0.04401
O3	57.81	0.73	-0.04	0.00	-3.56	0.0479982
N	21.14	-0.39	0.04	0.02	-0.03	0.0140067
NH	28.06	5.10	-0.77	0.06	-1.05	0.0150146
NO	35.99	0.96	-0.15	0.01	-3.00	0.0300061
N2	35.52	1.13	-0.20	0.01	-4.55	0.0280134
NO2	56.83	0.74	-0.14	0.01	-5.46	0.0460055
N2O	60.30	1.03	-0.19	0.01	-6.86	0.0440128
HCO	52.79	2.67	-0.39	0.02	-7.46	0.029018
HNO	52.26	3.11	-0.60	0.04	-7.45	0.031014
HNO2	79.10	1.67	-0.26	0.01	-15.05	0.0470134
CH4	85.81	11.26	-2.11	0.14	-26.42	0.0160425

Appendix C

Mass Model

C.1 Coefficients

fuel	LOX-RP1		LOX-LH2		N2O4/UDMH		HNO3UDMH	
stage	1st	2nd	1st	2nd	1st	2nd	1st	2nd
coefficients								
1	-8,1950E+01	-1,1620E-05	6,2689E+02	1,6681E-06	3,1560E+01	-0.000194273537046085	2,8988E-08	2,6163E-08
2	0.0893609530747904	-0.00570015659911688	3,2425E+00	1,1857E-05	0.00236730946985661	0.000993865604086473	-1,3457E-06	-1,3692E-06
3	0.000998594340766718	-0.0225535309432343	0.00229529158705205	7,7343E-05	-0.00470785613164307	0.00463240392314599	-6,6553E-06	-7,1403E-06
4	0.417245758182904	-0.0138982587896948	-1,8264E+00	0.000341818501167905	-0.170690692682507	-0.0279037396558479	4,0052E-06	3,6378E-06 5
	0.000245001345258739	-0.00354960150869133	-0.00706423036273235	0.00152951394194586	-0.00176911670925505	-0.000889227818436271	-0.000367172861936392	-0.000378130556291152
6	-0.000229569881467097	-1,0027E-05	0.000769407327727055	-8,1648E-05	0.000170592741558107	9,2767E-05	6,4221E-06	7,1639E-06
7	4,0498E+01	0.000335566981133333	-5,1096E+01	1,2701E-05	6,3863E+00	-0.000290040094085747	-7,5236E-08	8,7638E-09
8	0.169065981206201	-0.00592992790878616	-0.0989485190368861	0.000106678358694880	-0.00709430081220401	0.00729728157850815	-3,2880E-06	-1,7506E-06
9	-0.0896879471852889	0.0325608839191525	0.118837885078358	0.00414294572564383	-0.00544462533953012	-0.00101798289178512	-2,2455E-05	-7,9907E-07
10	0.515786756700506	0.000985245381280954	0.667705907713865	6,4741E-05	-1,0791E+00	-0.00110547385431454	-4,9213E-07	-6,0835E-08
11	2,4860E+00	-0.00274245499293817	-2,4416E+01	1,2253E-05	-4,2110E+00	0.000423441954764371	3,4995E-07	1,7698E-07
12	-0.0100822238382491	0.113283463723300	-0.00849258688627999	5,0077E-05	0.00984946827151924	0.00511230602755891	-1,1889E-05	-1,0569E-05
13	-0.0141522591746413	-0.00138118611900693	0.0589547664246698	0.000672418632659042	0.00834320049175984	-0.000782296413242881	7,4346E-05	3,3416E-05
14	-0.457219603400206	-0.00303799464795769	-0.0918085522942510	0.000137283139903056	0.212419952044731	0.00777138609444641	-8,4831E-07	-3,1813E-08
15	0.0472279089384427	0.00341559597331444	-0.0141341616479929	5,7610E-05	0.0710577503760508	0.0274353844715771	2,1656E-06	-9,0918E-07
16	-1,2541E+01	0.000409517754292141	-2,6154E+01	7,3922E-06	-4,0828E+00	-8,2882E-05	-2,4816E-07	-2,6235E-07
17	-0.0986261791132828	-0.0243083093240393	-0.0583497449471259	5,8307E-05	0.179192654168321	0.00314789173359682	-1,9101E-06	-1,9754E-06
18	-0.0299938746345677	0.0124698413753571	0.0635798040026045	0.00209149896281358	0.0315598896191666	0.00312410320860606	-7,1867E-05	-7,5840E-05
19	-8,4592E+00	0.00210221747810008	0.574536197166921	4,2608E-05	-0.145697598354420	0.00225196128098142	-4,3172E-07	-2,9367E-07
20	0.536952200341378	0.00430281247212983	0.028516880065172	6,7191E-05	0.459800901895970	-0.00563610960776709	-1,5425E-06	-1,6247E-06
21	6,3258E+00	0.00424864047974198	0.366866246669510	2,4780E-05	-1,8103E+00	0.000539965006378801	-5,3216E-07	-5,7787E-07
22	-0.0149995401850834	0.0317156767531257	0.00701850290440376	0.000170664248485905	0.0155565877787382	0.0344677523610142	-6,5859E-07	-8,1620E-08
23	-1,2490E-05	0.00291736007486406	9,4838E-05	2,8205E-06	0.000364361290754723	-8,5923E-05	-3,9732E-05	-4,5560E-05
24	0.000108651504834968	-3,3488E-05	9,9511E-06	1,2463E-05	4,5780E-05	-5,0500E-05	4,4212E-05	4,5039E-05
25	0.00260313064975771	-0.0102741675761780	0.00134728285324919	0.00024387753120903	-0.000778310267555661	0.000174122057002602	-5,5021E-05	-1,0127E-05
26	-2,4003E-05	-0.00505413469191897	-1,3039E-05	0.000300707597539030	-0.00186270531073504	-0.000981885185750412	-1,0903E-04	-6,2858E-05
27	-0.0061057770222184	-0.00715555172416038	-0.00294058614124470	-0.00103570558070426	-0.00586217151076694	-0.000251740846257811	-1,2696E-06	-1,1571E-06
28	8,2957E-07	9,8285E-06	5,4663E-07	1,5260E-07	-3,0089E-06	-3,1909E-06	2,3769E-07	-8,2402E-07

C.2 Sample Data

C.2.1 LOX - RP1

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Angara A-2	25.200	2.355	294	359	6,7	2,66	Lox/Kerosene	1	Russia
Angara Stage 1	500.000	40.000	8181,129	337	22,5	3,9	Lox/Kerosene	1	Russia
Angara UM	140.000	10.000	2094,7	338	25	2,9	Lox/Kerosene	1	Russia
Atlas A	81.647	7.230	1517,412	282	23,17	3,05	Lox/Kerosene	2	USA
Atlas Able	114.495	4.200	363,218	309	20,64	3,05	Lox/Kerosene	1	USA
Atlas Agena LV-3A	117.150	2.390	363,218	309	20,27	3,05	Lox/Kerosene	1	USA
Atlas Agena SLV-3	117.026	2.326	386,3	316	20,67	3,05	Lox/Kerosene	1	USA
Atlas Agena SLV-3A	142.000	3.700	386,3	316	20,67	3,05	Lox/Kerosene	1	USA
Atlas B	107.610	3.980	363,218	309	21,9	3,05	Lox/Kerosene	1	USA
Atlas CCB	306.914	22.461	4151,998	338	32,46	3,81	Lox/Kerosene	1	USA
Atlas Centaur LV-3C	117.350	3.700	363,218	309	18,3	3,05	Lox/Kerosene	1	USA
Atlas Centaur SLV-3C/D	128.500	4.000	386,3	316	22,2	3,05	Lox/Kerosene	1	USA
Atlas D	113.050	2.347	363,218	309	21,2	3,05	Lox/Kerosene	1	USA
Atlas E/F	117.826	4.926	386,3	316	20,12	3,05	Lox/Kerosene	1	USA
Atlas G/H/I	142.536	4.236	386,3	316	22,2	3,05	Lox/Kerosene	1	USA
Atlas II	161.995	6.095	386,3	316	24,9	3,05	Lox/Kerosene	1	USA
Atlas IIA	162.495	6.595	386,3	316	24,9	3,05	Lox/Kerosene	1	USA
Atlas IIAS	161.950	6.050	386,3	316	24,9	3,05	Lox/Kerosene	1	USA
Atlas IIIA	195.628	13.725	4148,722	337	29	3,05	Lox/Kerosene	1	USA/Russia
Atlas MA-2	3.050	3.050	1517,422	282	0	4,9	Lox/Kerosene	2	USA
Atlas MA-3	3.174	3.174	1644,96	290	0	4,9	Lox/Kerosene	2	USA
Atlas MA-5	3.646	3.646	1896,01	294	0	4,9	Lox/Kerosene	2	USA
Atlas MA-5A	4.187	4.187	2093,7	299	0	4,9	Lox/Kerosene	2	USA
Atlas MA-5AS	5.632	5.632	2093,7	299	0	4,9	Lox/Kerosene	2	USA
Black Colt	40.482	8.283	402,053	353	39,6	4,6	Lox/Kerosene	1	USA
CZ-NGLV-200	69.000	6.000	1340,17	336	25	2,25	Lox/Kerosene	1	China
CZ-NGLV-300	147.000	12.000	2680,35	336	26,3	3,35	Lox/Kerosene	2	China
CZ-NGLV-KO	60.000	7.000	588	340	13	3,35	Lox/Kerosene	4	China
Delta 3 - 1	104.377	6.822	1085,79	302	20	2,44	Lox/Kerosene	1	USA
Delta Thor ELT	84.067	4.059	1030,208	296	22,4	2,44	Lox/Kerosene	1	USA
Delta Thor LT	70.354	3.715	866,71	290	21,43	2,44	Lox/Kerosene	1	USA
Delta Thor RS27	84.368	4.360	1030,218	296	22,37	2,44	Lox/Kerosene	1	USA
Delta Thor TA	49.442	3.175	866,71	290	18,41	2,44	Lox/Kerosene	1	USA
Delta Thor XLT	101.700	5.690	1032,002	295	26,05	2,44	Lox/Kerosene	1	USA

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Delta Thor XLT-C	101.900	5.900	1054,195	302	26,05	2,44	Lox/Kerosene	1	USA
EKR Stage 1	6.390	1.564	107,333	250	8,29	0,88	Lox/Kerosene	1	Russia
Energia RCS	17.000	2.000	84,935	352	5,7	3,7	Lox/Kerosene	1	Russia
Energia Strap-on	355.000	35.000	7906,1	337	37,7	3,9	Lox/Kerosene	1	Ukraine
Europa-1	89.406	6.997	1672,671	282	18,75	3,05	Lox/Kerosene	2	UK
G-38 Navaho Booster	76.870	16.780	2048,158	282	27,43	2,37	Lox/Kerosene	3	USA
GR-1 Stage 1	80.000	6.000	1676	325	18,5	2,68	Lox/Kerosene	4	Russia
GR-1 Stage 2	25.000	2.100	441	340	9,15	2,68	Lox/Kerosene	1	Russia
GR-1 Stage 3	8.500	1.000	66,6	350	4,6	2,41	Lox/Kerosene	1	Russia
H-1-1	85.800	4.400	866,71	290	22	2,44	Lox/Kerosene	1	Japan
Jarvis-1	950.000	60.000	15481,26	304	35	8,38	Lox/Kerosene	2	USA
Jupiter	54.431	5.443	758,711	282	18,28	2,67	Lox/Kerosene	1	USA
Jupiter Cluster	226.716	27.205	3034,285	282	18,28	5,34	Lox/Kerosene	4	USA
Keldysh Bomber	100.000	29.500	980	285	28	3,6	Lox/Kerosene	1	USSR
Keldysh Sled	50.000	25.000	5880	285	14	3,6	Lox/Kerosene	6	USSR
Kistler Stage 1	250.000	20.500	5049,4	331	18,4	6,7	Lox/Kerosene	3	USA
Kistler Stage 2	131.000	13.100	1769,1	348	23,6	4,27	Lox/Kerosene	1	USA
KSR-3	5.000	700	122,5	280	9,6	1	Lox/Kerosene	1	Korea South
Kvant	210.000	15.000	3216	336	23	3,9	Lox/Kerosene	4	Russia
Kvant-1	74.000	7.000	1102,2	335	10,3	2,6	Lox/Kerosene	1	Russia
Liberty 1	17.000	2.240	245	270	10,2	1,58	Lox/Kerosene	1	USA
Luna 8K72-0	43.400	3.800	990	312	19	2,68	Lox/Kerosene	1	Russia
Luna 8K72-1	95.300	6.900	936,5	315	28	2,6	Lox/Kerosene	1	Russia
Luna 8K72-2	8.100	1.120	49	316	2,84	2,56	Lox/Kerosene	1	Russia
Molniya 8K78-0	43.400	3.800	995,3	314	19	2,68	Lox/Kerosene	1	Russia
Molniya 8K78-1	100.500	6.800	941	315	28	2,99	Lox/Kerosene	1	Russia
Molniya 8K78-2	24.300	2.000	294	330	2,84	2,56	Lox/Kerosene	1	Russia
Molniya 8K78-3	5.100	1.080	65,41	340	2,84	2,56	Lox/Kerosene	1	Russia
Molniya 8K78M-0	43.400	3.770	995,3	314	19	2,68	Lox/Kerosene	1	Russia
Molniya 8K78M-1	100.600	6.798	976,7	315	28	2,99	Lox/Kerosene	1	Russia
Molniya 8K78M-2	24.800	1.976	298,1	330	2,84	2,56	Lox/Kerosene	1	Russia
Molniya 8K78M-3	6.660	1.160	66,6	340	3,2	2,4	Lox/Kerosene	1	Russia
Molniya M-3	4.500	1.050	66,6	340	2,64	2,41	Lox/Kerosene	1	Russia
N-11 1963 - V	70.000	6.000	784	347	18	2,68	Lox/Kerosene	4	Russia
N-11 Block A	485.000	47.000	11960	330	20	6,8	Lox/Kerosene	8	Russia
N-11 Block B	192.000	18.000	1560	347	12	4,8	Lox/Kerosene	8	Russia
N-111 - A	192.000	20.000	2540	331	12	4,8	Lox/Kerosene	12	Russia
N-11GR - V	8.200	1.700	88,2	346	5	2,4	Lox/Kerosene	1	Russia
N1 1962 - A	1.384.000	117.000	39420	331	30	10	Lox/Kerosene	24	Russia
N1 1962 - B	506.000	50.000	13778	347	20	6,8	Lox/Kerosene	8	Russia
N1 1962 - V	193.000	16.000	1560	347	12	4,8	Lox/Kerosene	4	Russia
N1 1964 - A	1.942.000	192.000	49420	331	30	10	Lox/Kerosene	30	Russia

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
N1 1964 - B	506.000	50.000	13700	346	20	6,8	Lox/Kerosene	8	Russia
N1 1964 - V	193.000	16.000	1560	347	12	4,8	Lox/Kerosene	4	Russia
N1 Block A	1.880.000	130.000	50300	330	30,1	10,3	Lox/Kerosene	30	Russia
N1 Block B	560.700	55.700	14039,98	346	20,5	6,8	Lox/Kerosene	8	Russia
N1 Block D	18.200	3.500	83,3	349	5,7	2,9	Lox/Kerosene	1	Russia
N1 Block G	61.800	6.000	445,996	353	9,1	4,4	Lox/Kerosene	1	Russia
N1 Block V	188.700	13.700	1608	353	14,1	4,8	Lox/Kerosene	4	Russia
N1F 1965 - A	1.940.000	200.000	57560	331	30	10	Lox/Kerosene	30	Russia
N1F 1965 - B	590.000	60.000	15600	346	26	6,8	Lox/Kerosene	8	Russia
N1F 1965 - V	300.000	20.000	2450	347	16	4,8	Lox/Kerosene	4	Russia
N1F Block A	2.070.000	126.230	49675,5	331	30	10	Lox/Kerosene	30	Russia
N1F Block B	620.000	55.700	14040,004	346	20	6,8	Lox/Kerosene	8	Russia
N1F Block V	210.000	13.700	1608	353	12	4,8	Lox/Kerosene	4	Russia
N1M 1965 - A	3.800.000	375.000	85300	330	50	10	Lox/Kerosene	30	Russia
N1M 1965 - B	620.000	75.000	21960	346	30	6,8	Lox/Kerosene	8	Russia
N1M 1965 - V	355.000	20.000	2940	347	20	4,8	Lox/Kerosene	4	Russia
N1U 1965 - A	1.934.000	195.000	49420	331	30	10	Lox/Kerosene	30	Russia
N1U 1965 - B	506.000	50.000	13700	346	20	6,8	Lox/Kerosene	8	Russia
N1U 1965 - V	193.000	15.000	1560	347	12	4,8	Lox/Kerosene	4	Russia
N3 Block A	150.000	10.000	2694,37	336	12	4,8	Lox/Kerosene	7	Russia
Nova 2 F-1 Module	1.134.000	77.000	15121	304	28,7	10,1	Lox/Kerosene	2	USA
Nova 59-4-1	2.177.000	136.000	38697	295	23,2	15,5	Lox/Kerosene	5	USA
Nova 59-4-2	590.000	45.000	7560	310	16,2	5,2	Lox/Kerosene	1	USA
Nova 60-8-1	3.628.000	227.000	61928	304	41,2	17,4	Lox/Kerosene	8	USA
Nova 9L-1	2.721.000	181.000	68048	304	42,7	15,2	Lox/Kerosene	9	USA
Nova 9L-2	1.814.000	125.000	30243	304	35,1	13,7	Lox/Kerosene	4	USA
Nova A-1	1.342.000	73.000	30616	304	28	9,8	Lox/Kerosene	4	USA
Nova B-1	1.995.000	100.000	45914	304	28	12,2	Lox/Kerosene	6	USA
Nova DAC-1	4.213.000	194.000	97840	299	25,9	21,3	Lox/Kerosene	4	USA
Nova GD-B-1	9.070.000	635.000	142451	310	45,2	20,6	Lox/Kerosene	16	USA
Nova GD-F-1	9.977.000	680.000	148031	335	54,3	18,3	Lox/Kerosene	4	USA
Nova GD-J-1	8.617.000	680.000	124044	330	36,6	42,4	Lox/Kerosene	4	USA
Nova MM 1B-1	7.430.000	454.000	125770	304	46,6	20	Lox/Kerosene	14	USA
Nova MM 1C-1	8.943.000	590.000	161710	304	49,7	21	Lox/Kerosene	18	USA
Nova NASA-1	2.268.000	113.000	45914	296	31,4	14,6	Lox/Kerosene	6	USA
Nova NASA-2	499.000	35.000	7640	304	12,5	9,8	Lox/Kerosene	1	USA
Pathfinder	100.000	30.000	833	350			Lox/Kerosene	1	USA
Project 921-1	306.000	20.000	5406,141	330	29	4,5	Lox/Kerosene	4	China
Proton 11S824	13.360	1.800	83,3	346	5,5	3,7	Lox/Kerosene	1	Russia
Proton 11S824F	16.900	1.800	85,02	352	5,5	3,7	Lox/Kerosene	1	Russia

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Proton 11S824M	14.000	1.830	85,02	352	5,5	3,7	Lox/Kerosene	1	Russia
Proton 11S86	17.550	2.300	85,02	352	7,1	3,7	Lox/Kerosene	1	Russia
Proton 11S861	17.300	2.300	85,02	352	7,1	3,7	Lox/Kerosene	1	Russia
Proton 11S861-01	18.650	2.650	83,45	361	7,1	3,7	Lox/Kerosene	1	Russia
Proton 17S40	14.600	3.300	85,02	352	7,1	3,7	Lox/Kerosene	1	Russia
R-3	69.000	5.480	1370	285	27,1	2,8	Lox/Kerosene	1	Russia
R-7 8K71-0	43.100	3.500	970	306	19	2,68	Lox/Kerosene	1	Russia
R-7 8K71-1	95.300	7.500	912	308	28	2,99	Lox/Kerosene	1	Russia
R-7A 8K74-0	42.800	3.200	995,3	313	19	2,68	Lox/Kerosene	1	Russia
R-7A 8K74-1	101.700	7.300	940,4	315	28	2,99	Lox/Kerosene	1	Russia
R-9A Stage 1	61.600	4.000	1595,83	311	14,79	2,68	Lox/Kerosene	1	Russia
R-9A Stage 2	15.900	2.500	304	330	9,4	2,68	Lox/Kerosene	1	Russia
RLA-1	800.000	45.000	11700	337	30	6	Lox/Kerosene	1	USSR
RLA-2	150.000	10.000	2940	349	12	6	Lox/Kerosene	1	USSR
Sanger-Bredt	99.773	19.955	1428,8	306	28	2,32	Lox/Kerosene	1	Germany
Saturn I	432.681	45.267	7582,1	289	24,48	6,52	Lox/Kerosene	8	USA
Saturn IB	448.648	41.594	8241,763	296	24,48	6,52	Lox/Kerosene	8	USA
Saturn IB-11	505.953	48.545	8241,763	296	30,48	6,52	Lox/Kerosene	8	USA
Saturn IB-15	478.103	45.873	8241,763	296	27,43	6,52	Lox/Kerosene	8	USA
Saturn IC	2.286.217	135.218	38703,16	304	42,06	10,06	Lox/Kerosene	5	USA
Saturn IC C-3B	1.160.787	72.549	22954,798	304	34,41	10,06	Lox/Kerosene	5	USA
Saturn IC C-4B	1.450.984	90.686	30606,397	304	33,46	10,06	Lox/Kerosene	5	USA
Saturn IC C-5A	2.217.285	131.495	38257,99	304	42,87	10,06	Lox/Kerosene	5	USA
Saturn IC-Flat Bulkhead	2.278.800	127.000	38717,6	304	36,55	10,06	Lox/Kerosene	5	USA
Saturn LCB-Lox/RP-1	1.360.800	108.900	20306,6	275	48,78	6,61	Lox/Kerosene	1	USA
Saturn MLV 5-23L-0	1.180.000	80.000	15484	304	51	6,61	Lox/Kerosene	2	USA
Saturn MS-IC-1	2.694.703	169.719	45948,754	310	48,15	10,06	Lox/Kerosene	5	USA
Saturn MS-IC-1A	2.694.700	160.600	46443,3	304	48,16	10,06	Lox/Kerosene	6	USA
Saturn MS-IC-23(L)	2.694.000	169.000	38703,16	304	48,15	10,06	Lox/Kerosene	5	USA
Saturn MS-IC-25(S)	3.223.800	211.900	45925,5	304	54,7	10,06	Lox/Kerosene	5	USA
Saturn MS-IC-3B	2.718.800	178.700	45925,5	304	48,15	10,06	Lox/Kerosene	5	USA
Saturn MS-IC-4(S)B	2.875.500	154.000	38717,6	304	50,58	10,06	Lox/Kerosene	5	USA
Saturn MS-IC/260	2.721.600	181.400	38717,6	304	42,05	10,06	Lox/Kerosene	5	USA
Saturn MS-LRB-23(L)	874.100	18.100	15486,6	304	47,85	6,61	Lox/Kerosene	2	USA
Saturn S-1B-4 H-1	444.700	37.600	4120,7	296	24,48	6,52	Lox/Kerosene	4	USA
Saturn S-1B-2	725.491	68.014	13339,956	304	34,5	8,25	Lox/Kerosene	2	USA
Saturn S-1B-4	1.813.729	108.823	30962,526	304	28,96	10,06	Lox/Kerosene	4	USA
Saturn S-1B-A	448.600	41.600	9045,6	296	24,48	6,52	Lox/Kerosene	8	USA
Saturn S-1C-8	3.627.500	181.400	61925	304	48,79	12,19	Lox/Kerosene	8	USA
Saturn S-1C-TLB	952.500	68.000	15532,7	304	36,36	6,61	Lox/Kerosene	2	USA
Saturn S-ID Booster	67.100	67.100	30962,5	304	42,07	10,06	Lox/Kerosene	4	USA
Saturn S-ID Sustainer	2.219.100	68.100	7740,3	304	42,07	10,06	Lox/Kerosene	1	USA

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Saturn S-ID Sustainer-1	2.627.600	84.400	7740,3	304	48,16	10,06	Lox/Kerosene	1	USA
Sea Dragon-1	11.600.000	1.300.000	356900	290	68	21,5	Lox/Kerosene	1	USA
Sea Horse-1	40.000	5.000	755	285	15,85	3	Lox/Kerosene	1	USA
Soyuz 11A511-0	43.000	3.550	994,3	314	19,6	2,68	Lox/Kerosene	1	Russia
Soyuz 11A511-1	100.500	6.500	977,7	315	27,8	2,95	Lox/Kerosene	1	Russia
Soyuz 11A511-2	25.400	2.700	294	330	6,74	2,66	Lox/Kerosene	1	Russia
Soyuz 11A511U-0	44.500	3.784	994,3	310	19,6	2,68	Lox/Kerosene	1	Russia
Soyuz 11A511U-1	105.400	6.875	997,091	311	27,8	2,95	Lox/Kerosene	1	Russia
Soyuz 11A511U-2	25.200	2.355	298,1	330	6,74	2,66	Lox/Kerosene	1	Russia
Soyuz 11A511U2-0	42.810	3.550	996,3	314	19,6	2,68	Lox/Kerosene	1	Russia
Soyuz 11A511U2-1	101.160	6.500	1011	319	27,8	2,95	Lox/Kerosene	1	Russia
Soyuz 11A511U2-2	25.400	2.400	298,1	330	6,74	2,66	Lox/Kerosene	1	Russia
Soyuz M-0	42.200	3.000	872	336	20	2,68	Lox/Kerosene	1	Russia
Soyuz M-1	100.500	6.000	872	336	28	2,95	Lox/Kerosene	1	Russia
Soyuz M-3	5.000	1.000	20	360	2,7	2,4	Lox/Kerosene	1	Russia
Soyuz ST-0	44.400	3.810	1021,097	310	19,6	2,68	Lox/Kerosene	1	Russia
Soyuz ST-1	105.400	6.875	999,601	311	27,8	2,95	Lox/Kerosene	1	Russia
Soyuz ST-2	25.200	2.355	294	359	6,74	2,66	Lox/Kerosene	1	Russia
Sputnik 8A91-0	43.000	3.400	970	310	19	2,68	Lox/Kerosene	1	Russia
Sputnik 8A91-1	95.000	7.100	804	315	28	2,99	Lox/Kerosene	1	Russia
Sputnik 8K71PS-0	43.000	3.400	970	306	19	2,68	Lox/Kerosene	1	Russia
Sputnik 8K71PS-1	94.000	7.495	912	308	28	2,99	Lox/Kerosene	1	Russia
Super-Jupiter	317.600	19.600	7538,3	290	22	5,23	Lox/Kerosene	4	USA
Super-Jupiter-2	96.700	7.200	1884,592	290	18	3,05	Lox/Kerosene	1	USA
Super-Jupiter-3	29.400	1.900	412,8	308	7	3,05	Lox/Kerosene	1	USA
Superraket Block A	1.150.000	100.000	33656	310	27	9	Lox/Kerosene	66	Russia
Thor DM-19	49.340	3.125	758,711	282	18,42	2,44	Lox/Kerosene	1	USA
Thor DM-21	48.354	2.948	760,643	285	18,41	2,44	Lox/Kerosene	1	USA
Titan 1-1	76.203	4.000	1467,908	290	16	3,05	Lox/Kerosene	2	USA
Titan 1-2	28.939	1.725	355,863	308	9,8	2,26	Lox/Kerosene	1	USA
Titan Cluster	150.000	10.000	2940	130	16	6,1	Lox/Kerosene	4	USA
UR-500MK-1	137.000	9.300	1637	331	20	4,15	Lox/Kerosene	1	USSR
UR-500MK-2	161.000	11.000	1860	346	20	4,15	Lox/Kerosene	1	USSR
UR-700M-1	8.000.000	750.000	204027	337	35	9	Lox/Kerosene	32	Russia
UR-700M-2	5.250.000	500.000	102013,6	337	30	12,5	Lox/Kerosene	16	Russia
Vanguard	7.661	811	134,792	270	12,2	1,14	Lox/Kerosene	1	USA
VKS-R Sled	100.000						Lox/Kerosene		Russia
Voskhod 11A57-0	43.400	3.800	995,3	314	19	2,68	Lox/Kerosene	1	Russia
Voskhod 11A57-1	100.500	6.800	941	315	28	2,99	Lox/Kerosene	1	Russia
Voskhod 11A57-2	24.300	2.000	294	330	2,84	2,56	Lox/Kerosene	1	Russia
Vostok 8A92-0	43.300	3.700	995,3	313	19	2,68	Lox/Kerosene	1	Russia

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Vostok 8A92-1	100.400	6.800	940,4	315	28	2,99	Lox/Kerosene	1	Russia
Vostok 8A92-2	7.775	1.440	54,9	326	2,84	2,56	Lox/Kerosene	1	Russia
Vostok 8A92M-0	43.300	3.700	995,3	313	19	2,68	Lox/Kerosene	1	Russia
Vostok 8A92M-1	101.000	6.500	940,4	315	28	2,99	Lox/Kerosene	1	Russia
Vostok 8A92M-2	8.100	1.100	54,52	326	2,84	2,56	Lox/Kerosene	1	Russia
Vostok 8K72-0	43.300	3.710	970	312	19	2,68	Lox/Kerosene	1	Russia
Vostok 8K72-1	100.400	6.800	912	315	28	2,99	Lox/Kerosene	1	Russia
Vostok 8K72-2	7.775	1.440	49,42	316	2,84	2,56	Lox/Kerosene	1	Russia
Vostok 8K72K-0	43.300	3.710	970	313	19	2,68	Lox/Kerosene	1	Russia
Vostok 8K72K-1	100.400	6.800	912	315	28	2,99	Lox/Kerosene	1	Russia
Vostok 8K72K-2	7.775	1.440	54,52	326	2,84	2,56	Lox/Kerosene	1	Russia
Vulkan 0	355.000	35.000	7891,01	336	39	3,9	Lox/Kerosene	1	
X-34	21.800	8.200	268,927	310	17,68		Lox/Kerosene	1	USA
X-34A-1	29.500	6.700	386,3	316	21,95	3,05	Lox/Kerosene	1	USA
X-34A-2	4.500	300	44,1	348	4	2	Lox/Kerosene	1	USA
YaKhR-2 Strap-on	110.000	8.250	2393,724	311	27	3,33	Lox/Kerosene	6	Russia
Zenit-1	354.300	28.600	8181,129	337	32,9	3,9	Lox/Kerosene	1	Ukraine
Zenit-2	90.600	9.000	912	349	11,5	3,9	Lox/Kerosene	1	Ukraine
Zenit-3	17.300	2.720	84,92	352	5,6	3,7	Lox/Kerosene	1	Ukraine

C.2.2 LOX - LH2

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Albatros Carrier Aircraft	1,250,000	210,000	7840	455	91	7	Lox/LH2	4	Russia
Albatros Raketoplan	320,000	82,000	1960	455	49	7	Lox/LH2	4	Russia
ALS	780,000	60,000	15477.159	435	60	8.7	Lox/LH2	6	USA
Angara KVRB	23,300	3,500	73.5	461	8.6	4.1	Lox/LH2	1	Russia
Angara Stage 2	75,000	9,000	1960	455	16	3.9	Lox/LH2	1	Russia
Ares Stage 1	787,700	64,200	8705.99	453	43	8.7	Lox/LH2	3	USA
Ares Stage 2	172,000	13,200	1113	465	16	10	Lox/LH2	1	USA
Ariane 1-3	9,687	1,457	61.674	443	10.23	2.66	Lox/LH2	1	Germany
Ariane 2-3	12,000	1,600	62.703	446	11.53	2.66	Lox/LH2	1	Germany
Ariane 4-3	12,310	1,570	62.703	446	11.85	2.66	Lox/LH2	1	Germany
Ariane 5 EPC	186,000	12,700	1114	434	30.5	5.46	Lox/LH2	1	France
Ariane 5 ESC A	16,500	2,100	64.7	446	9	5.46	Lox/LH2	1	Germany
Ariane 5 ESC B	27,500	3,400	153.9	467	10	5.46	Lox/LH2	1	Germany
Ariane 5-1	170,800	12,700	1114	430	30.5	5.46	Lox/LH2	1	France
Astro-1	302,183	32,558	8820	410	29		Lox/LH2	1	USA
Astro-2	89,290	14,000	1170	425	20.7		Lox/LH2	1	USA
Beta	450,000	40,000	5736	460	40	7.65	Lox/LH2	13	Germany

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Cargo LV Stage 1	1,093,330	88,449	10441.061	452	64.27	8.4	Lox/LH2	5	USA
Cargo LV Stage 2	227,036	19,343	2442.071	451	22.74	8.4	Lox/LH2	2	USA
Centaur B-X	19,138	2,358	186.797	470	10.1	3.05	Lox/LH2	2	USA
Centaur C	15,600	1,996	133.448	425	9.14	3.05	Lox/LH2	2	USA
Centaur C-X	19,138	2,358	110.8	450	10.1	3.05	Lox/LH2	1	USA
Centaur D/E	16,258	2,631	131.222	444	9.6	3.05	Lox/LH2	2	USA
Centaur G	23,880	2,775	146.8	444	9	4.33	Lox/LH2	2	USA
Centaur G Prime	19,501	3,000	146.8	444	8.87	4.33	Lox/LH2	2	USA
Centaur G STS	16,327	2,600	146.8	444	5.95	4.33	Lox/LH2	2	USA
Centaur I	15,600	1,700	146.8	444	9.15	3.05	Lox/LH2	2	USA
Centaur II	18,833	2,053	146.8	444	10.1	3.05	Lox/LH2	2	USA
Centaur IIA	19,073	2,293	185.012	449	10.1	3.05	Lox/LH2	2	USA
Centaur IIIA	18,710	1,905	99.155	451	10	3.05	Lox/LH2	1	USA
Centaur IIIB	22,960	2,130	198.319	451	11.68	3.05	Lox/LH2	2	USA
Centaur V1	22,825	2,026	99.194	451	12.68	3.05	Lox/LH2	1	USA
Centaur V2	23,050	2,250	198.398	451	12.68	3.05	Lox/LH2	2	USA
Chang Cheng 1	57,000	30,000	490	230	24	4.5	Lox/LH2	1	China
CLV Stage 2	160,000	16,000	2333	465	32	5.5	Lox/LH2	1	USA
CZ-3-3	10,500	2,000	44.1	425	7.48	2.25	Lox/LH2	4	China
CZ-3A-3	21,000	2,800	156	440	12.38	3	Lox/LH2	2	China
CZ-NGLV-500	175,000	17,000	1399.99	432	31	5	Lox/LH2	2	China
CZ-NGLV-HO	26,000	3,100	156	448	12	3.35	Lox/LH2	2	China
DAC Helios ISI-1	1,333,000	120,000	49807	455	22.9	24.4	Lox/LH2	4	USA
DAC Helios-1	1,660,000	149,000	52926	410	16.2	24.4	Lox/LH2	4	USA
DC-I	470,000	36,000	6860	425	38.72	9.15	Lox/LH2	1	USA
DC-X	16,320	7,200	262.8	373	11.89	3.05	Lox/LH2	4	USA
DC-Y	84,000	6,000	1051.2	373	20	5	Lox/LH2	16	USA
Delta 3 - 2	19,300	2,476	110.03	462	8.8	2.44	Lox/LH2	1	USA
Delta 4 - 2	24,170	2,850	110.05	462	12	2.44	Lox/LH2	1	USA
Delta 4H - 2	30,710	3,490	110.05	462	12	2.44	Lox/LH2	1	USA
Delta RS-68	226,400	26,760	3312.755	420	40.8	5.1	Lox/LH2	1	USA
Energia Core	905,000	85,000	7848.124	453	58.77	7.75	Lox/LH2	4	Russia
Energia EUS	77,000	7,000	1962.026	455	16.47	5.7	Lox/LH2	1	Russia
Energia M	272,000	28,000	1960	455	20	7.7	Lox/LH2	1	Russia
GSLV-3	14,600	2,200	75.05	460	8.72	2.8	Lox/LH2	1	India
H-1-2	10,600	1,800	102.9	450	10.32	2.49	Lox/LH2	1	Japan
H-2-1	98,100	11,900	1077.996	446	28	4	Lox/LH2	1	Japan
H-2-2	16,700	2,700	121.5	452	10.6	4	Lox/LH2	1	Japan
H-2A LRB	117,000	17,800	2196	440	36.7	4	Lox/LH2	2	Japan

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
H-2A-1	113,600	13,600	1098	440	37.2	4	Lox/LH2	1	Japan
H-2A-2	19,600	3,000	137	447	9.2	4	Lox/LH2	1	Japan
Helios A-1	200,000	10,000	6660	400	8.8	6.7	Lox/LH2	4	USA
Helios B-1	236,000	14,000	7786	400	8.8	6.7	Lox/LH2	5	USA
Helios C-1	395,000	18,000	12450	400	11	8.5	Lox/LH2	8	USA
HIMES	13,600	3,100	274	452	13.6	2.4	Lox/LH2	2	Japan
Hyperion Booster	394,625	18,144	13700	457	12	8.54	Lox/LH2	4	
Hyperion SSTO	450,000	44,000	7840	459	18	10	Lox/LH2	1	USA
Interim HOTOL	250,000	33,100	7840	455	36.45	10	Lox/LH2	4	UK/USSR
Jarvis-2	145,000	15,000	1031.983	425	10	8.38	Lox/LH2	1	USA
Magnum Core	830,000	70,000	9112.378	453	53	8.4	Lox/LH2	4	USA
McDonnell-Douglas ILRV	215,000	95,254	16242.823		53		Lox/LH2	4	USA
McDonnell-Douglas ILRV Drop Tanks	1,084,086	40,823			61.9		Lox/LH2		USA
MLLV	5,352,400	317,400	71171.7	455	78.63	21.86	Lox/LH2	12	USA
Mustard 1	141,043	24,036	2150	405	30	4	Lox/LH2	1	UK
Mustard 2	142,184	24,943	2150	405	30	4	Lox/LH2	1	UK
N1 Block R	23,000	4,300	73.5	440	8.7	4.1	Lox/LH2	1	Russia
N1 Block S	58,000	8,000	392	440	16	6.7	Lox/LH2	1	Russia
N1 Block Sr	77,900	11,500	147.88	441	16.5	5.2	Lox/LH2	2	Russia
N1 Block V-II	805,000	115,000	7840	440	50	10	Lox/LH2	4	Russia
N1 Block V-III	325,000	35,000	2350	440	27	8	Lox/LH2	6	Russia
NLS Core	815,732	44,757	7150	430	52	8.7	Lox/LH2	2	USA
NLS HLV	833,732	62,757	14310	425	52	8.7	Lox/LH2	4	USA
NLS Semistage	36,000	36,000	14310	425	9	8.7	Lox/LH2	4	USA
Nova 4 J-2	771,000	63,000	3559	420	38.1	10.1	Lox/LH2	4	USA
Nova 59-4-3	168,000	18,000	1421	420	14	5.2	Lox/LH2	4	USA
Nova 59-4-4	68,000	9,000	353	420	11.9	3	Lox/LH2	1	USA
Nova 60-8-2	680,000	54,000	10669	428	35.1	11.6	Lox/LH2	2	USA
Nova 60-8-3	227,000	23,000	892	425	30.5	6.7	Lox/LH2	1	USA
Nova 9L-3	363,000	36,000	5334	420	21.3	10.1	Lox/LH2	6	USA
Nova 9L-4	118,000	14,000	1775	420	20.7	6.7	Lox/LH2	2	USA
Nova A-2	403,000	29,000	5334	420	23.8	9.8	Lox/LH2	4	USA
Nova A-3	40,000	4,000	313	427	13.1	5.8	Lox/LH2	5	USA
Nova B-2	614,000	38,000	10228	420	23.8	12.2	Lox/LH2	7	USA
Nova B-3	63,000	5,000	451	427	18.3	6.7	Lox/LH2	7	USA
Nova DAC 2-1	4,943,000	311,000	102293	410	82.3	21.3	Lox/LH2	15	USA
Nova DAC 2-2	1,633,000	113,000	13346	426	21.3	21.3	Lox/LH2	4	USA
Nova DAC ISI-1	3,084,000	142,000	70146	350	30.5	21.3	Lox/LH2	14	USA
Nova DAC ISI-2	1,696,000	107,000	33754	455	30.5	21.3	Lox/LH2	8	USA
Nova DAC-2	2,268,000	143,000	26683	426	27.4	21.3	Lox/LH2	4	USA
Nova GD-B-2	998,000	75,000	10669	428	27.1	20.6	Lox/LH2	5	USA
Nova GD-E-2	4,535,000	363,000	26683	428	47.9	20.6	Lox/LH2	4	USA
Nova GD-F-2	1,497,000	91,000	13346	428	29.9	18.3	Lox/LH2	2	USA
Nova GD-H-0	295,000	295,000	109402	410	21.2	25.9	Lox/LH2	4	USA
Nova GD-H-1	8,526,000	172,000	30685	428	73.8	25.9	Lox/LH2	1	USA

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Nova GD-J-2	1,270,000	91,000	13346	428	38	19.5	Lox/LH2	2	USA
Nova MM 14A-2	3,401,000	295,000	33352	428	28.8	18.3	Lox/LH2	5	USA
Nova MM 14B-2	2,721,000	245,000	26683	428	50.5	18.3	Lox/LH2	4	USA
Nova MM 1B-2	1,361,000	122,000	13346	428	45.4	20	Lox/LH2	2	USA
Nova MM 1C-2	2,041,000	163,000	20015	428	49.7	18.3	Lox/LH2	3	USA
Nova MM 24G-1	5,040,000	317,000	95261	451	49.8	21.6	Lox/LH2	18	USA
Nova MM 24G-2	1,043,000	136,000	10228	451	28.8	18.3	Lox/LH2	2	USA
Nova MM 33-1	10,489,000	626,000	156876	443	75.9	24.4	Lox/LH2	24	USA
Nova MM 34-0	227,000	227,000	122749	439	18.2	27.4	Lox/LH2	1	USA
Nova MM 34-1	10,126,000	544,000	30685	439	84.8	24.4	Lox/LH2	4	USA
Nova MM S10E-1	10,317,000	635,000	160672	454	65.7	24.4	Lox/LH2	24	USA
Nova MM S10E-2	10,302,000	639,000	160672	454	54.8	21.3	Lox/LH2	30	USA
Nova MM S10R-1	10,470,000	816,000	160672	454	69.5	24.4	Lox/LH2	24	USA
Nova MM S10R-2	10,502,000	839,000	160672	454	56.6	21.3	Lox/LH2	30	USA
Nova MM T10EE-1-1	4,944,000	317,000	96379	454	57.5	21.3	Lox/LH2	18	USA
Nova MM T10EE-1-2	1,125,000	136,000	11032	454	23.3	21.3	Lox/LH2	2	USA
Nova MM T10RE-1-1	4,979,000	454,000	96379	454	57.8	21.3	Lox/LH2	18	USA
Nova MM T10RR-2-1	9,089,000	680,000	176686	304	43.5	21.3	Lox/LH2	24	USA
Nova MM T10RR-2-2	2,041,000	181,000	20015	455	24.4	21.3	Lox/LH2	1	USA
Nova MM T10RR-3-1	5,517,000	680,000	109000	454	61.3	21.3	Lox/LH2	18	USA
Nova MM T10RR-3-2	1,229,000	136,000	12052	454	21.8	21.3	Lox/LH2	2	USA
Nova NASA-3	227,000	21,000	2667	420	11	9.8	Lox/LH2	4	USA
OOST	7,982,000	431,000	123191	410	85.4	21.3	Lox/LH2	4	USA
OOST ISI	5,125,000	292,000	85386	455	57.9	21.3	Lox/LH2	15	USA
Pegasus Tanks	20,000	3,700	0	459	18.9	4.57	Lox/LH2	0	USA
Pegasus VTOVL	1,250,000	148,000	23947	459	34.76	10	Lox/LH2	1	USA
Project 921-2	57,000	7,000	490	440	10	4.5	Lox/LH2	1	China
Proton KM-4	19,500	3,000	73.579	461	7	3.8	Lox/LH2	1	Russia
Rombus	5,102,041	306,175	101900	455	28.96	16	Lox/LH2	1	USA
Rombus Tank	107,501	18,143	0	455	30.49	7.62	Lox/LH2	0	USA
ROOST	10,898,000	608,000	165447	410	106.7	21.3	Lox/LH2	4	USA
ROOST ISI	6,218,000	435,000	101842	455	67.1	21.3	Lox/LH2	18	USA
Sanger I-1	79,000	10,000	1912	430	30	3.35	Lox/LH2	3	Germany
Sanger I-2	31,500	18,000	637	430	20.5	2.2	Lox/LH2	1	Germany
Sanger II-2	112,000	32,600	1280.003	490	27.6	5.5	Lox/LH2	1	Germany
SASSTO	97,976	6,668	1558.1	464	18.8	6.6	Lox/LH2	1	USA
Saturn II	490,778	39,048	5165.79	421	24.84	10.06	Lox/LH2	5	USA
Saturn II C-5A	384,057	31,740	4446.648	420	21.39	10.06	Lox/LH2	5	USA
Saturn II-INT-17	495,000	48,000	9713.4	450	24.84	10.06	Lox/LH2	7	USA
Saturn II-SL	490,952	44,240	5500.765	390	24.84	10.06	Lox/LH2	5	USA
Saturn IV	50,576	5,217	400.346	410	12.19	5.49	Lox/LH2	6	USA

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Saturn IVB	119,920	13,311	1031.983	425	17.8	6.61	Lox/LH2	1	USA
Saturn IVB (S-IB)	118,800	12,900	1031.6	421	17.8	6.61	Lox/LH2	1	USA
Saturn IVB (S-V)	119,900	13,300	1031.6	421	17.8	6.61	Lox/LH2	1	USA
Saturn IVB C-3B	52,144	6,801	889.325	420	15.4	5.59	Lox/LH2	1	USA
Saturn IVB C-5A	102,249	11,563	889.325	420	19	5.59	Lox/LH2	1	USA
Saturn IVB-A	118,800	12,900	956.1	421	17.8	6.61	Lox/LH2	1	USA
Saturn MS-II-1	521,447	49,877	5165.79	425	25.88	10.06	Lox/LH2	5	USA
Saturn MS-II-1-J-2T-200K	521,447	49,877	4446.648	435	25.88	10.06	Lox/LH2	5	USA
Saturn MS-II-1-J-2T-250K	521,447	49,877	5558.31	441	25.88	10.06	Lox/LH2	5	USA
Saturn MS-II-1A	600,800	60,000	6380.941	421	29.59	10.06	Lox/LH2	7	USA
Saturn MS-II-2	600,788	59,844	7003.477	451	29.59	10.06	Lox/LH2	5	USA
Saturn MS-II-3B	643,200	53,500	12455.4	447	29.56	10.06	Lox/LH2	5	USA
Saturn MS-II-4(S)B	494,100	42,300	5169	421	24.84	10.06	Lox/LH2	5	USA
Saturn MS-IVB-1	120,500	13,900	1031.6	421	18.08	6.61	Lox/LH2	1	USA
Saturn MS-IVB-1A	179,200	20,400	1031.6	421	22.82	6.61	Lox/LH2	1	USA
Saturn MS-IVB-2	176,600	17,800	1401.3	447	23.12	6.61	Lox/LH2	1	USA
Saturn MS-IVB-3B	179,200	20,400	1778.9	447	22.82	6.61	Lox/LH2	1	USA
Saturn MS-IVB-4(S)B	118,400	14,100	1031.6	421	17.79	6.61	Lox/LH2	1	USA
Saturn MS-IVB-x	195,000	36,300	912	421	18.08	6.61	Lox/LH2	1	USA
Saturn S-II	100,000	14,000	3557.313	420	22.56	6.6	Lox/LH2	4	USA
Saturn S-II-4	294,731	24,938	3557.313	420	16.76	10.06	Lox/LH2	4	USA
Saturn S-II-8	770,835	63,480	8265.26	425	42.68	10.06	Lox/LH2	8	USA
Saturn S-II-C3	204,044	24,938	3557.313	420	21.3	8.25	Lox/LH2	4	USA
Saturn S-IVC	204,044	22,671	1031.983	425	22.86	6.61	Lox/LH2	1	USA
Sea Dragon-2	5,900,000	530,000	62270	320	44	21.5	Lox/LH2	1	USA
Sea Horse-2	15,000	2,000	147	425	10	3	Lox/LH2	1	USA
Shuttle DC-3-1	799,537	131,519	10290	442	61.89	5.08	Lox/LH2	4	USA
Shuttle DC-3-2	199,238	54,422	2940	459	37.4	4.45	Lox/LH2	1	USA
Shuttle FR-3-1	2,169,691	234,467	30088.43	442	71.8	11.19	Lox/LH2	15	USA
Shuttle FR-3-2	388,189	130,159	4549.049	459	54.57	7.46	Lox/LH2	2	USA
Shuttle H33-1	1,489,717	224,431	24973.399	442	74.7	10	Lox/LH2	12	USA
Shuttle H33-2	474,199	100,153	7079.45	459	47.87	8.07	Lox/LH2	3	USA
Shuttle HCR-1	1,634,467	304,535	29135.625	442	70.79	10	Lox/LH2	14	USA
Shuttle HCR-2	342,948	129,329	4719.636	459	48.26	7.5	Lox/LH2	2	USA
Shuttle LCR-1	1,512,381	286,621	27054.517	442	75.15	10	Lox/LH2	13	USA
Shuttle LCR-2	322,449	120,816	4719.636	459	45	8	Lox/LH2	2	USA
Shuttle LRB	350,000	52,000	10318.106	435	45.54	5.5	Lox/LH2	4	USA
Shuttle LS A-1	1,225,668	162,494	26076.588	442	67.07	10	Lox/LH2	13	USA
Shuttle LS A-2	406,344	104,891	6823.56	459	49.7	8	Lox/LH2	3	USA
Shuttle LS200-1	1,730,803	133,514	27422.001	455	47.71	4.57	Lox/LH2	9	USA
Shuttle MDC-A-1	1,242,630	220,254	18956	442	59.45	8	Lox/LH2	10	USA

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Shuttle MDC-A-2	335,601	73,442	3070.4	459	32.62	7	Lox/LH2	2	USA
Shuttle NAR A-1	1,641,723	273,469	28130.189	442	85.37	9.88	Lox/LH2	11	USA
Shuttle NAR A-2	395,011	121,542	5256.893	459	61.59	6.71	Lox/LH2	2	USA
Shuttle Orbiter	99,318	99,117	6834.303	455	37.24	4.9	Lox/LH2	3	USA
Shuttle R134C-1	1,764,039	351,538	28824.774	442	81.4	10.37	Lox/LH2	12	USA
Shuttle R134C-2	424,449	138,910	4804.13	459	64.02	4.57	Lox/LH2	2	USA
Shuttle R134G-1	1,600,000		28824.774	442	78.35	10.37	Lox/LH2	12	USA
Shuttle R134G-2	380,000		4804.13	459	64.02	4.57	Lox/LH2	2	USA
Shuttle SERV-1	2,040,816	226,757	31980.515	455	20.27	18.29	Lox/LH2	1	USA
Shuttle Super Lightweight Tank	748,000	27,000		453	47	8.4	Lox/LH2	0	USA
Shuttle Tank	750,975	29,930	0	455	46.88	8.4	Lox/LH2	0	USA
SLS Stage A	59,000	6,000	889.325	424	15	4.28	Lox/LH2	1	USA
SLS Stage B	160,000	11,200	1778.651	424	25	7.62	Lox/LH2	2	USA
SLS Stage C	825,000	58,000	10673.322	428	53	7.62	Lox/LH2	2	USA
Spacemaster-1	1,224,490	203,342	28082.538	442	57.93	8	Lox/LH2	14	USA
Spacemaster-2	362,812	80,000	4549.049	459	48.78	6.71	Lox/LH2	2	USA
Starlifter	42,630	19,955	5217	455	25	15.85	Lox/LH2	3	USA
Starlifter Tank	257,778	12,971	0	455	25	8	Lox/LH2	0	USA
Titan 5	500,000	45,000	4457.1	425	60	6	Lox/LH2	1	USA
Titan C-2	57,400	6,000	666	403	19.5	4.07	Lox/LH2	1	USA
UR-700M-3	2,000,000	200,000	11700	455	16	12.5	Lox/LH2	6	Russia
Venturestar	991,000	89,300	15413.89	455	39.7	39	Lox/LH2	7	USA
VTOHL 45t	1,158,192	96,650	18643.471	467	53.5	34.3	Lox/LH2	5	USA
VTOHL 9t	687,503	71,901	11186.092	467	58.5	36	Lox/LH2	3	USA
VTOVL 150t	4,093,761	419,154	63387.802	485	26.5	29.7	Lox/LH2	17	USA
Vulkan 1	800,000	80,000	7450	452	52	7.75	Lox/LH2	4	Russia
Vulkan Blok V	142,000	15,000	411	460	16.5	6.7	Lox/LH2	1	Russia
X-33	123,800	28,600	2284	439	20.4	20.7	Lox/LH2	2	USA

C.2.3 N2O4 - UDMH

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
8K94	9,800	900	147	330	8.4	1.6	N2O4/UDMH	1	Ukraine
Angara Briz M	6,565	1,600	19.6	326	2.61	2.5	N2O4/UDMH	1	Russia
Ariane 1-1	160,030	13,750	2771.94	281	18.4	3.8	N2O4/UDMH	4	France
Ariane 1-2	37,130	3,625	720.965	296	11.5	2.6	N2O4/UDMH	1	France
Ariane 2-1	160,030	13,750	2879.997	281	18.4	3.8	N2O4/UDMH	4	France
Ariane 2-2	37,130	3,625	804.998	296	11.5	2.6	N2O4/UDMH	1	France
Ariane 4-1	245,900	17,900	3034.1	278	23.6	3.8	N2O4/UDMH	4	France
Ariane 4L	43,772	4,493	752.003	278	19	2.22	N2O4/UDMH	1	France
Burlak Stage 1	18,000	1,800	357.9	320	10.5	1.6	N2O4/UDMH	1	Russia
Burlak Stage 2	9,400	900	98	330	5.5	1.6	N2O4/UDMH	1	Russia

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
CZ-1C-3	6,400	1,000	107	287	4.5	1.8	N2O4/UDMH	1	China
CZ-1D-2	14,850	2,650	294	287	5.35	2.25	N2O4/UDMH	1	China
CZ-2C-1	153,000	10,000	3295	291	20.52	3.35	N2O4/UDMH	4	China
CZ-2C-2	39,000	4,000	761.9	295	7.5	3.35	N2O4/UDMH	1	China
CZ-2C/SD-2	55,000	5,000	761.9	298	10	3.35	N2O4/UDMH	1	China
CZ-2E(A)-0	80,000	8,000	1632.8	291	27	2.25	N2O4/UDMH	2	China
CZ-2E-0	41,000	3,200	816.285	291	15.33	2.25	N2O4/UDMH	1	China
CZ-2E-1	196,500	9,500	3265.143	289	23.7	3.35	N2O4/UDMH	4	China
CZ-2E-2	91,500	5,500	831.005	298	15.52	3.35	N2O4/UDMH	1	China
CZ-3-1	151,000	9,000	3000	289	20.22	3.35	N2O4/UDMH	4	China
CZ-3-2	39,000	4,000	761.9	295	9.71	3.35	N2O4/UDMH	1	China
CZ-3A-1	179,000	9,000	3265.143	289	23.08	3.35	N2O4/UDMH	4	China
CZ-3A-2	33,600	4,000	831.005	297	11.53	3.35	N2O4/UDMH	1	China
CZ-4A-1	192,700	9,500	3265.143	289	24.66	3.35	N2O4/UDMH	4	China
CZ-4A-2	39,550	4,000	831.005	295	10.41	3.35	N2O4/UDMH	1	China
CZ-4A-3	15,150	1,000	100.81	303	1.92	2.9	N2O4/UDMH	2	China
Diamant B-1	21,500	3,500	396.522	251	14.08	1.4	N2O4/UDMH	4	France
Diamant-1	14,685	1,946	301.55	251	9.76	1.34	N2O4/UDMH	4	France
Diamant-2	2,815	540	120.082	255	4.57	0.8	N2O4/UDMH	1	France
Europa-2	11,894	2,099	274	277	5.7	2	N2O4/UDMH	4	France
FB-1-1	150,400	10,000	3000	289	20.1	3.35	N2O4/UDMH	4	China
FB-1-2	38,300	3,500	761.9	295	7.4	3.35	N2O4/UDMH	1	China
Fregat	6,535	1,100	19.6	327	1.5	3.35	N2O4/UDMH	1	Russia
GSLV-0	45,600	5,600	735	281	19.7	2.1	N2O4/UDMH	1	India
GSLV-2	42,900	5,400	725.015	295	11.6	2.8	N2O4/UDMH	1	India
Ikar	3,164	820	2.94	326	2.56	2.72	N2O4/UDMH	1	Russia
LCLV-1	3,253,246	396,013	56368	267	39.33	12.2	N2O4/UDMH	1	USA
LCLV-2	798,389	95,092	9793.303	300	25.91	9.15	N2O4/UDMH	1	USA
LCLV-3	161,304	19,518	2028.044	306	15.24	5.79	N2O4/UDMH	1	USA
Martlet 4L-2	243	46	20.5	300	1.85	0.42	N2O4/UDMH	1	Canada
Martlet 4L-3	58	9	5.39	300	0.69	0.42	N2O4/UDMH	1	Canada
Proton K-1	450,510	31,100	10470.158	316	21.2	4.15	N2O4/UDMH	6	Russia
Proton K-2	167,828	11,715	2399.216	327	14	4.15	N2O4/UDMH	4	Russia
Proton K-3	50,747	4,185	630.17	325	6.5	4.15	N2O4/UDMH	1	Russia
Proton KM-1	450,400	31,000	10532	317	21	7.4	N2O4/UDMH	6	Russia
Proton KM-3	50,300	3,700	613.8	327	6.9	4.15	N2O4/UDMH	1	Russia
Proton KM-4 Briz M	22,170	2,370	19.6	326	2.61	2.5	N2O4/UDMH	1	Russia
Proton-2	134,900	13,180	2399.19	325	10.9	4.15	N2O4/UDMH	4	Russia
PSLV-2	45,800	5,300	725.015	293	12.5	2.8	N2O4/UDMH	1	India
R-36-1	122,300	6,400	2651.7	301	18.9	3	N2O4/UDMH	1	Ukraine
R-36-2	49,300	3,700	940.4	317	9.4	3	N2O4/UDMH	1	Ukraine

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
R-36-0-1	125,000	8,500	2640	301	18.9	3	N2O4/UDMH	1	Ukraine
R-36-0-2	48,000	5,000	955.991	317	9.4	3	N2O4/UDMH	1	Ukraine
R-36M2-1	210,800	15,000	4520	318	22.3	3	N2O4/UDMH	1	Ukraine
R-36M2-2	49,300	5,000	755	340	5.7	3	N2O4/UDMH	1	Ukraine
R-36M2-3	8,200	2,000	18.6	317	1	3	N2O4/UDMH	1	Ukraine
R-56 Block A	1,162,000	162,000	1608	316	36	6.5	N2O4/UDMH	16	Ukraine
R-56 Block B	213,000	13,000	1980	325	12	6.5	N2O4/UDMH	1	Ukraine
R-56 Block O	32,000	2,000	544.2	327	7	4	N2O4/UDMH	1	Ukraine
Rokot-3 Briz	6,565	1,600	19.6	326	1.3	2.5	N2O4/UDMH	1	Russia
Saturn LCB-Storable-140	1,995,800	199,600	28553	267	57.16	6.61	N2O4/UDMH	1	USA
Saturn LCB-Storable-250	1,451,500	108,900	21232.3	267	40.7	6.61	N2O4/UDMH	1	USA
Saturn PFL	2,057,400	205,400	28553	267	33.57	10.06	N2O4/UDMH	1	USA
Tsyklon 2-1	122,300	6,400	2640	301	18.9	3	N2O4/UDMH	1	Ukraine
Tsyklon 2-2	49,300	3,700	940.4	317	9.4	3	N2O4/UDMH	1	Ukraine
Tsyklon 2-3	3,200	400	77.96	317	2.5	2	N2O4/UDMH	1	Russia
Tsyklon 3-1	127,000	8,300	3032	301	18.75	3	N2O4/UDMH	1	Ukraine
Tsyklon 3-2	53,300	4,800	941	318	10.08	3	N2O4/UDMH	1	Ukraine
Tsyklon 3-3	4,600	1,407	78.708	317	2.58	2.25	N2O4/UDMH	1	Russia
UR-100 Stage 1	32,799	3,000	868.8	310	10	1.6	N2O4/UDMH	4	Russia
UR-100 Stage 2	8,611	900	131.448	325	6.69	2	N2O4/UDMH	1	Russia
UR-100N MIRV Bus	1,100	725	4.9	200	0.5	2.5	N2O4/UDMH	1	Russia
UR-100N Stage 1N	77,150	5,695	1778.661	310	17.2	2.5	N2O4/UDMH	4	Russia
UR-100N Stage 2N	12,195	1,485	214.27	322	3.9	2.5	N2O4/UDMH	1	Russia
UR-200 Stage 1	107,500	6,600	2235	311	20	3	N2O4/UDMH	1	Russia
UR-200 Stage 2	25,810	2,400	612.9	322	10	2.2	N2O4/UDMH	1	Russia
UR-500 Monoblok Stage 1M	367,360	25,560	8989.7	310	30	6.2	N2O4/UDMH	4	USSR
UR-500 Polyblock Stage 1P	378,900	26,900	8989.7	310	21.2	4.1	N2O4/UDMH	4	USSR
UR-500 Stage 2	137,700	16,000	3183.2	320	10.9	4.1	N2O4/UDMH	4	USSR
UR-500 Stage 3	23,940	2,180	613.11	320	3.7	4.1	N2O4/UDMH	1	USSR
UR-700-1	3,210,000	222,100	40364	322	53.6	17.6	N2O4/UDMH	6	Russia
UR-700-2	1,072,000	75,600	20182	322	34.2	8.3	N2O4/UDMH	3	Russia
UR-700-3	399,400	26,000	5139.6	328	20.83	4.15	N2O4/UDMH	3	Russia
UR-700-4	33,500	1,900	131.4	326	6.2	2.7	N2O4/UDMH	1	Russia

C.2.4 HN03 - UDMH

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Able	2,268	816	34.3	270	6.57	0.84	Nitric acid/UDMH	1	USA
Able-Star	4,470	590	35.088	278	5.88	1.4	Nitric acid/UDMH	1	USA
Agenda A	3,790	885	68.95	276	4.73	1.52	Nitric acid/UDMH	1	USA
Agenda B	7,167	867	71.166	285	7.09	1.52	Nitric acid/UDMH	1	USA

Stage	Gross Mass-kg	Empty Mass-kg	Thrust(vac)-kN	Isp(vac)-sec	Length-m	Diameter-m	Propellants	No Engines	Country
Agena D	6,821	673	71.166	300	7.09	1.52	Nitric acid/UDMH	1	USA
CZ-1-1	64,100	4,100	1224.575	268	17.84	2.25	Nitric acid/UDMH	4	China
CZ-1-2	15,000	2,700	306.143	275	5.35	2.25	Nitric acid/UDMH	1	China
Delta 104	4,472	590	35.098	278	5.88	1.4	Nitric acid/UDMH	1	USA
Delta A	2,164	694	33.803	271	5.36	0.84	Nitric acid/UDMH	1	USA
Delta D	2,693	545	33.695	278	5.58	0.84	Nitric acid/UDMH	1	USA
Delta E	6,009	785	35.098	278	6.28	1.4	Nitric acid/UDMH	1	USA
Delta F	5,629	784	41.364	280	6.28	1.4	Nitric acid/UDMH	1	USA
Kosmos-1	87,200	5,300	1740.023	292	22.4	2.4	Nitric acid/UDMH	1	Ukraine
Kosmos-2	20,135	1,435	156	303	6	2.4	Nitric acid/UDMH	1	Ukraine
R-16 Stage 1	110,000	10,000	2609.5	289	16	3	Nitric acid/UDMH	1	
R-16 Stage 2	27,000	3,000	882.1	293	13	2.4	Nitric acid/UDMH	1	
R-26 Stage 1	60,300	3,000	1451	310		3	Nitric acid/UDMH	1	
R-26 Stage 2	25,200	2,500	428.06	308		2.4	Nitric acid/UDMH	1	
Taepodong 1-1	25,649	3,733	577.337	255	12	1.8	Nitric acid/UDMH	4	Korea North
Taepodong 1-2	7,505	1,456	144.334	255	12	0.96	Nitric acid/UDMH	1	Korea North
Von Braun 1956-1	1,100,000	140,000	28257.8	257	22.2	11.7	Nitric acid/UDMH	29	Germany
Von Braun 1956-2	154,000	14,000	3254.23	297	16.9	7.8	Nitric acid/UDMH	8	Germany
Von Braun 1956-3	12,400	2,100	406.77	297	2.9	2.14	Nitric acid/UDMH	1	Germany

Bibliography

- [1] M. Di Giacinto. "Dispense del corso di Propulsione Aerospaziale. (Italian) [Aerospace Propulsion course notes]". In: *Università di Roma La Sapienza, dispense del corso di Propulsione Aerospaziale - Lesson 18b* (2018), pp. 11–12. URL: http://dma.ing.uniroma1.it/users/aeroprop_c1/testolez18b.pdf.
- [2] Dieter K. Huzel and David H. Huang. *Design of Liquid Propellant Rocket Engines*. Springer Aerospace Technology, special publication (SP). 19710019929. 1971, p. 91. URL: <https://ntrs.nasa.gov/citations/19710019929>.
- [3] Bonnie J. McBride and Sanford Gordon. *Computer Program for Calculation of Complex Chemical Equilibrium Compositions and Applications II. Users Manual and Program Description*. NASA Lewis Research Center Cleveland, OH United States. NASA Reference Publication (RP). 19960044559. 1996, pp. 9–10. URL: <https://ntrs.nasa.gov/citations/19960044559>.
- [4] Edouardo Perez. "Soyuz User Manual". In: *Arianespace publications* (2012), pp. 1–244. URL: <https://www.arianespace.com/wp-content/uploads/2015/09/Soyuz-Users-Manual-March-2012.pdf>.
- [5] National Institute of Standards and Technology. U.S. Department of Commerce. *NIST Chemistry Web-Book*. URL: <https://webbook.nist.gov/chemistry/>.
- [6] Mark Wade. *Astronautix, Ariane 1-3 H8*. URL: <http://www.astronautix.com/a/ariane1-3h8.html>.
- [7] Mark Wade. *Astronautix, Ariane 44LP*. URL: <http://www.astronautix.com/a/ariane44lp.html>.
- [8] Mark Wade. *Astronautix, Ariane 5-1 H155*. URL: <http://www.astronautix.com/a/ariane5-1h155.html>.
- [9] Mark Wade. *Astronautix, Atlas Able*. URL: <http://www.astronautix.com/a/atlasable.html>.
- [10] Mark Wade. *Astronautix, Chang Zheng 2C*. URL: <http://www.astronautix.com/c/changzheng2c.html>.
- [11] Mark Wade. *Astronautix, Falcon 9*. URL: <http://www.astronautix.com/f/falcon9.html>.
- [12] Mark Wade. *Astronautix, Kosmos 65*. URL: <http://www.astronautix.com/k/kosmos65s3.html>.
- [13] Mark Wade. *Astronautix, Soyuz-FG*. URL: <http://www.astronautix.com/s/soyuz-fg.html>.
- [14] Ten-See Wang. "Thermophysics Characterization of Kerosene Combustion". In: *JOURNAL OF THERMOPHYSICS AND HEAT TRANSFER* 15.2 (2001), pp. 140–141. doi: <https://doi.org/10.2514/2.6602>.